

Task value inside the AI time-horizon frontier

Tom Harris*

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Abstract

METR evaluates AI agents by the length of a task they can autonomously complete. This paper puts that measure on an economic scale. Mapping human duration estimates from the Anthropic Economic Index to 18,796 O*NET work tasks, I estimate the share of the US wage bill that falls inside the AI capability frontier. Weighting each task by its implied success probability, I find that tasks carrying 60 per cent of the wage bill are expected to succeed under the latest frontier model, whereas tasks actively delegated to Claude account for 44 per cent cumulatively. This capability baseline hinges on two definitions. First, success follows METR’s definition: an agent completes a well-specified software task that is scored automatically. On real freelance projects that a client must accept, agents succeed far less often than METR’s curves predict. Second, the baseline treats all non-physical work as potentially automatable. Counting only occupations that can be performed remotely, 29 per cent of all US employment is in jobs where at least half of the work is expected to succeed, and no occupation reaches 90 per cent. For now, the capability frontier covers parts of many jobs rather than the whole of a few.

1 Introduction

METR measures the performance of AI agents by the maximum length of the tasks they can autonomously complete, expressed as the time a human would require for the same work (Kwa et al., 2025). On METR’s evaluation suite, this time horizon has doubled roughly every 4.2 months since 2023 (METR, 2026). Over the same period, the Anthropic Economic Index (AEI) has tracked which work tasks users delegate to Claude and estimated how long those pieces of work would take a person without AI (Handa et al., 2025; Anthropic, 2026). This paper places both measures on a single scale, US work tasks weighted by their share of the wage bill, to measure two quantities: the share of the wage bill in tasks a frontier agent is expected to complete, and the share in tasks that workers actually delegate to an AI system.

Macroeconomic scenario models of AI and the labour market, such as the framework developed at Anthropic by Korinek et al. (2026), rely on assumed values for both of these quantities. Their “extreme” 2030 scenario, for example, assumes an affected task mass (m_t) of 50 per cent of all tasks and an automation share (ψ_t) of 0.90, meaning that nine in ten AI-performed tasks are automated

*London School of Economics; t.m.harris@lse.ac.uk. Work in progress: this draft is preliminary and incomplete. Comments and corrections are very welcome. Please do not cite or circulate without permission.

rather than augmented. Across the values chosen for these parameters, the framework produces outcomes that range from negligible disruption to 17.9 per cent cognitive unemployment, 15.4 per cent annual GDP growth and a labour share of income of 45.2 per cent. Establishing an empirical baseline for the parameters is therefore critical.

To measure capability, I use METR’s fitted logistic curves, which map an agent’s completion probability to the logarithm of a task’s human time. Rather than counting tasks below a threshold such as the 50 or 80 per cent horizon, I calculate the probability of success for each task given its length and sum the wage bill weighted by that probability. This expected-success share is a continuous measure of capability.

I anchor the estimates on two models: Claude Opus 4.6, the most capable model inside METR’s 16-hour reliability range, and Mythos Preview (early), a pre-release checkpoint that is the latest measured frontier but sits above that range. Under Claude Opus 4.6 the expected-success share is 52 per cent of the US wage bill; under Mythos Preview (early) it is 60 per cent. Tasks delegated to Claude represent 25 per cent of the wage bill in the most recent AEI May 2026 release, and 44 per cent cumulatively across all AEI releases. Capability and use do not nest: workers use AI on pieces of longer tasks that the agent cannot complete alone, while many short tasks that the curve expects to succeed remain unused.

These shares use METR’s definition of success, in which an agent completes a well-specified task that is scored automatically. Stricter standards give much lower success. METR’s curves predict that the same frontier agents would complete 30–51 per cent of the freelance projects in the Remote Labor Index, where clients accepted 1.2–4.2 per cent (Mazeika et al., 2025). Converting this gap into a share of the wage bill depends on assumptions the benchmark cannot test, so I report the conversion only as an illustration (Section 5.4).

The closest work is the January 2026 Anthropic Economic Index report (Appel et al., 2026), which combines per-task success rates classified from Claude conversations with task time shares into an “effective AI coverage” of occupations, and estimates usage-based time horizons for comparison with METR’s. Massenkoff and McCrory (2026) build an observed-exposure measure from the same usage data, which Korinek et al. (2026) use to anchor their scenarios. This paper differs in using a capability curve measured on a benchmark rather than success classified from usage, in weighting by the wage bill, and in tracing the measure across frontier models since 2023. It builds on task-based models of technical change (Autor et al., 2003; Acemoglu and Restrepo, 2018) and on measures of occupational exposure to AI (Brynjolfsson et al., 2018; Webb, 2020; Felten et al., 2021; Eloundou et al., 2024), which Acemoglu (2025) turns into a macroeconomic estimate by combining exposure with the share of exposed tasks that is cost-effective to automate (Svanberg et al., 2024).

Translating benchmarks to the economy requires two extrapolations: AEI’s duration estimates for delegated pieces of work stand in for the length of employer-defined tasks, and success curves estimated on software engineering tasks are applied to all non-physical work. AEI’s own success data suggest that the second is the weaker link (Section 3). I bound the estimates with a compound bootstrap and a set of robustness checks, and report how far each definition moves the result.

2 Data

2.1 METR time horizons

Figure 1 reproduces METR’s headline capability estimates. Each point represents a model’s 50 per cent time horizon. METR flags measurements above 16 hours as unreliable because its current suite has few long tasks. Mythos Preview (early) is the only frontier model above this threshold, and Claude Opus 4.6 is the last model measured below it; I report both throughout.

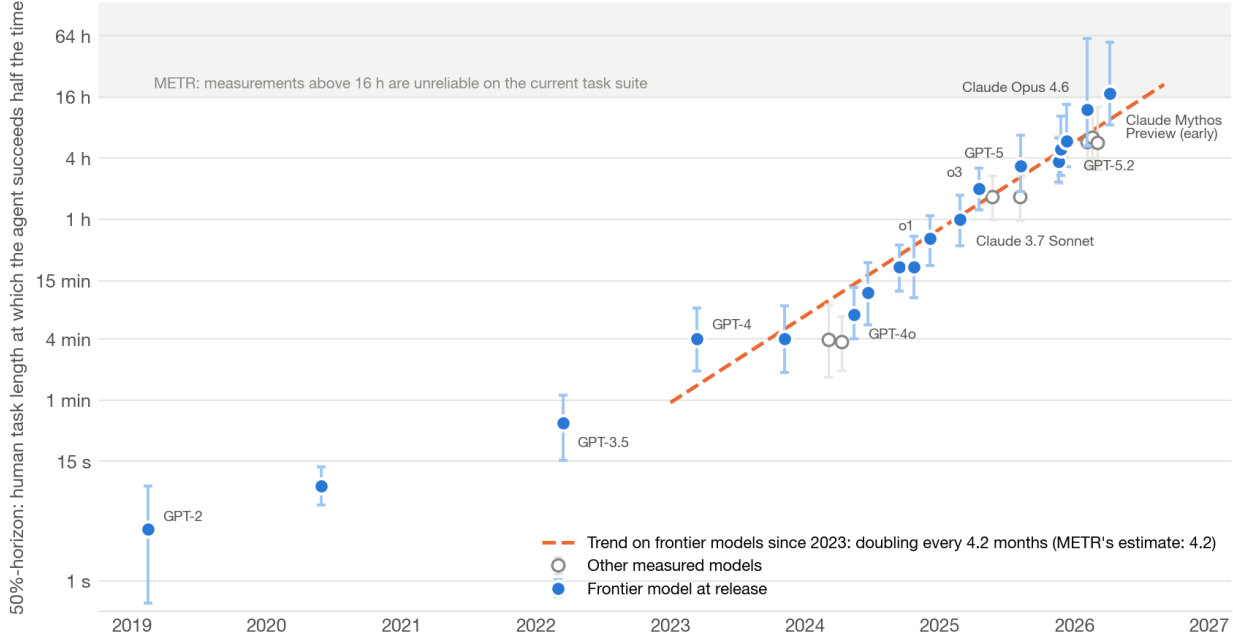


Figure 1: METR’s 50 per cent time horizons, reproduced from METR’s published estimates (METR, 2026). Bars are METR’s published 95 per cent confidence intervals. Filled points: frontier model at release. Dashed line: exponential trend fitted to frontier models released from January 2023 with horizons below 16 hours. Grey band: horizons METR describes as unreliable on the current suite.

METR’s evaluation suite is specialised, covering 228 computer-based tasks across 79 families (Appendix Table 7). Most come from HCAST, a set of software engineering, machine learning, cybersecurity and research tasks with completion times benchmarked on human contractors (Rein et al., 2025). To construct the expected-success shares, I rebuilt every run from the raw per-task success counts and refit the logistic curves using METR’s methodology. The refit reproduces METR’s published horizons to within 0.02 per cent for every model (Appendix Table 8) and supplies the coefficients for the continuous success curve.

2.2 The Anthropoc Economic Index

The AEI reports the share of Claude conversations mapped to specific O*NET tasks, alongside an AI-generated estimate of the human time required to complete the delegated work without AI (Handa et al., 2025; Anthropic, 2026). For task durations, I rely on the June 2026 release because it maps tasks by their exact O*NET IDs; earlier releases are retained for robustness checks.

Durations are directly observed for 2,814 tasks, representing 26 per cent of the wage bill. For the

remaining tasks, I impute the duration using the geometric mean of observed durations among tasks sharing an O*NET detailed work activity (see Appendix A for fallback levels and hold-out validation). The resulting wage-weighted median task duration across the economy is 3.1 hours.

The duration of a task depends on how the work reaches Claude. The June release is the first whose Claude.ai data include Cowork, Anthropic’s agentic desktop product, and for the same task its durations are 1.6 times those in the February release, while API durations barely change. For the same task in the same month, API durations are 0.26 to 0.50 times Claude.ai durations (Appendix K). The earlier releases give an expected-success share of 64 per cent (Table 13).

2.3 Wage weights from O*NET and OEWS

I map each O*NET task to its parent occupation and assign it a proportional share of the occupation’s total wage bill based on OEWS May 2025 employment and mean annual wages (U.S. Bureau of Labor Statistics, 2026; National Center for O*NET Development, 2026). The share is proportional to importance over the occupation’s core tasks, using O*NET’s own core/supplemental classification. Summed over all 18,796 tasks, this yields \$10.72 trillion across 153.4 million jobs. Hours-implied weights, OEWS annualisation conventions and admitting the supplemental tasks all leave the aggregate within a point of the baseline (Table 13).

2.4 Tasks, work activities and the physical filter

Which work a software agent could do is a matter of definition. The baseline uses the most direct definition O*NET allows, and it is likely an overestimate. A task is classified as physical if *any* of its underlying O*NET detailed work activities maps to a physical generalized work activity (handling objects, controlling machines, operating vehicles, assisting and caring for others, and similar). It is classified as non-physical only if *none* do. Table 1 illustrates this mapping.

The strict rule reflects the premise that the expected-success share measures the completion of a task as defined: a software agent cannot complete a mixed task unless an employer unbundles it and reassigns the physical part. Of the 18,796 tasks in the economy, 4,819 are entirely physical and 657 mix physical and non-physical activities. Together these carry 22 per cent of the wage bill and are assigned a success probability of zero.

The rule records what a task involves, not where it is done. O*NET files “sell tickets and collect fees from customers” under selling and “measure felled logs” under estimating quantities, so amusement attendants and log graders pass the filter. Adding the public-facing and inspecting work activities to the physical set moves only 4.6 per cent of the wage bill and leaves both occupations in place. Setting is recorded only at the level of the occupation, so as a bound I also apply the rule of Dingel and Neiman (2020), rebuilt on O*NET 30.2, which excludes every task in occupations that cannot be done remotely. Non-physical tasks in such occupations carry 30 per cent of the wage bill. Because Dingel and Neiman count school teaching as remote (it could be done at home in 2020), I also exclude occupations that require close physical contact (Appendix J). Figure 2 shows the range: the expected-success share under Mythos Preview (early) runs from 32 per cent with both exclusions (35 per cent with Dingel–Neiman alone) to 63 per cent with a lenient filter that excludes a task only if every activity is physical; the Eloundou et al. (2024) exposure labels give 49 per cent. The definition of automatable work moves the share by more than a model generation. The aggregate results below use the baseline filter; the results for individual occupations in Section 5.1

Table 1: From task to work activity: the O*NET mapping for the highest-importance task in each class for Registered Nurses, and for the highest-importance physical task of Construction Laborers. A task is physical if any detailed work activity it maps to sits under a physical generalized work activity.

Task (occupation)	Detailed work activities → generalized work activity	Counted as
Record patients’ medical information and vital signs. (Registered Nurses)	Record patient medical histories. → Documenting/Recording Information	non-physical
Administer medications to patients and monitor patients for reactions or side effects. (Registered Nurses)	Monitor patient conditions during treatments, procedures, or activities. → Monitoring Processes, Materials, or Surroundings Administer non-intravenous medications. → Assisting and Caring for Others (physical)	physical
Provide health care, first aid, immunizations, or assistance in convalescence or rehabilitation in locations such as schools, hospitals, or industry. (Registered Nurses)	Immunize patients. → Assisting and Caring for Others (physical) Treat acute illnesses, infections, or injuries. → Assisting and Caring for Others (physical)	physical
Tend pumps, compressors, or generators to provide power for tools, machinery, or equipment or to heat or move materials, such as asphalt. (Construction Laborers)	Operate pumps or compressors. → Controlling Machines and Processes (physical)	physical

Table 2: Task universes. Shares are of the \$10.72 trillion wage bill attached to all tasks.

Universe	Tasks	Share of wage bill
All O*NET tasks in occupations with OEWS employment and wages	18,796	100.0%
Non-physical tasks (no physical work activity)	13,320	78.3%
Tasks with an observed AEI duration (June 2026 AEI release, Apr and May 2026)	2,814	25.6%
Computer and Mathematical occupations (in-domain check, SOC 15)	770	5.9%

use both exclusions.

3 Measures

Let D_t denote task t ’s duration in human hours, and $P_d(D)$ the fitted success curve of the frontier model at date d . The expected-success share is:

$$S_d = \frac{\sum_t V_t P_d(D_t) \mathbf{1}[t \text{ non-physical}]}{\sum_t V_t}.$$

The threshold shares, F_d^{50} and F_d^{80} , replace the continuous probability $P_d(D_t)$ with an indicator equal to one if D_t is below the 50 or 80 per cent horizon. Because S_d integrates the full curve, it gives partial credit to tasks beyond the threshold and discounts shorter tasks by their residual failure rate. It is a measure of capability: it does not account for the cost of checking the agent’s work or for the friction of reassigning standing responsibilities to an agent.

Evaluating S_d across the economy requires extrapolating METR’s success curves out of domain. The core assumption is that, at a given human duration, an agent’s probability of completing a

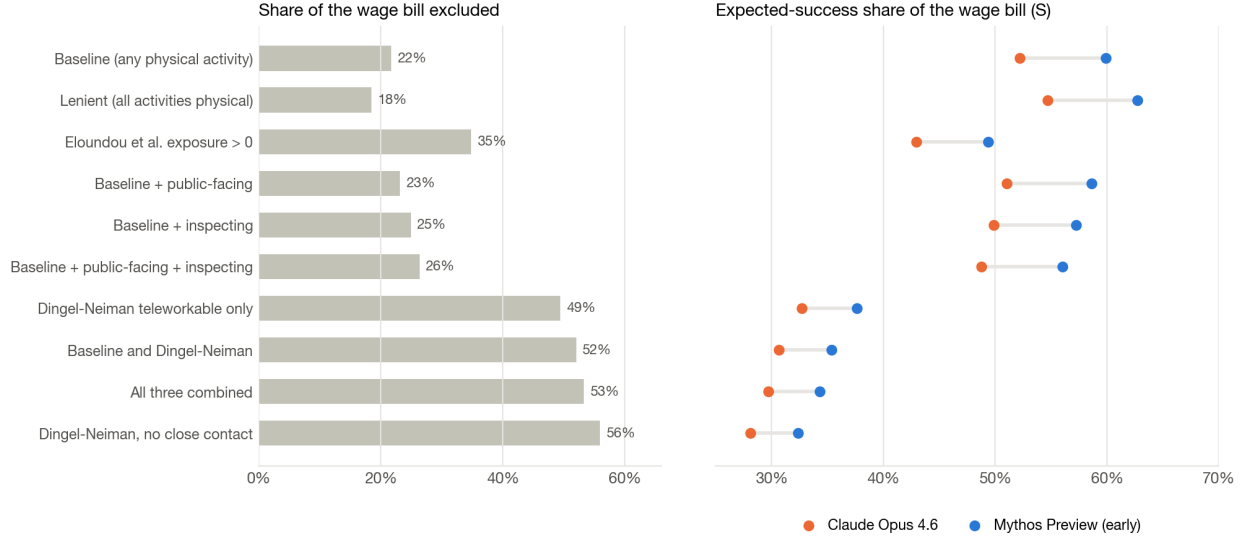


Figure 2: Ten rules for the excluded set. Left: share of the wage bill excluded. Right: expected-success share of the wage bill under both frontier models. Baseline: a task is excluded if any of its work activities is physical. Lenient: excluded only if every work activity is physical. Eloundou et al.: excluded if the exposure label is zero. Public-facing, inspecting: tasks with a work activity under generalized work activities 4.A.4.a.8 or 4.A.1.b.2 are also excluded. Dingel-Neiman: every task of an occupation that cannot be done entirely at home is excluded, using their published rule rebuilt on O*NET 30.2 (Appendix J). No close contact: also excludes occupations whose O*NET physical-proximity rating is 4 or more (at arm’s length or closer), such as school teaching, which Dingel-Neiman count as remote because it can be delivered through a screen.

paralegal’s non-physical task equals the probability predicted for a software engineering task of the same length.

Within METR’s suite, task length predicts average success well. For individual frontier models, the weighted R^2 of task success rates on the pooled curve is 0.24 for Mythos Preview (early) and 0.26 for Claude Opus 4.6. With four to six runs per task, binomial noise caps the attainable R^2 at roughly 0.57, so length accounts for about 43 per cent of the explainable variation (Appendix D). Because S_d is linear in P_d , it requires only that the curve capture the conditional mean of success at a given length, not the outcome of each task. The curve is also stable to the composition of the suite: dropping any single task family moves Mythos Preview (early)’s expected-success share by no more than two points (59 to 61 per cent).

Outside the suite, the evidence for the duration gradient is weaker. Two AEI releases report, for each task, the share of conversations that Claude classified as successful alongside their human time. On Claude.ai this share barely changes with duration: across tasks, the fall in the log-odds of success per doubling of human time is between -0.04 and 0.04 , against 0.41 to 0.66 for the METR curves of the Claude models that bracket that traffic. On the API it falls, but more slowly, at 0.12 to 0.21 per doubling within occupations and work activities (Appendix I). These labels are conditional on a user choosing to delegate the work and allow the user to iterate, so they do not measure autonomous completion, and success that varies within a workflow rather than with task length is consistent with a “jagged” frontier (Dell’Acqua et al., 2026). But they give no support to the gradient across tasks that the METR curve imposes. I keep METR’s curves, whose gradient is measured under controlled conditions, and treat the variation of success across tasks as the least secure part of the measure. METR’s comparison across benchmarks points the same way: horizons

on visual computer-use tasks are 40 to 100 times shorter than on software (Kwa and Cheng, 2025).

Every headline table includes an in-domain check restricted to Computer and Mathematical occupations (SOC 15, 5.9 per cent of the wage bill), where the capability assumption is least strained.

4 Results

Table 3 presents the baseline estimates under both frontier models. Moving from Claude Opus 4.6 to Mythos Preview (early) increases the expected-success share of the US wage bill from 52 to 60 per cent. A compound bootstrap that resamples the METR curve, adds measurement noise to AEI durations and re-runs the imputation yields a 95 per cent interval of 52 to 66 per cent under the latest frontier. The interval reflects sampling uncertainty only.

The share of the wage bill in tasks shorter than the 80 per cent horizon moves sharply, from 9 to 31 per cent as the frontier advances. The 50 per cent threshold share, by contrast, is saturated, covering nearly the entire non-physical wage bill.

Table 3: Headline numbers, May 2026. “Non-physical” = no physical work activity. Durations imputed where unobserved. With w_N the non-physical share of the wage bill and $\bar{P}_N$, $\bar{P}_P$ the wage-weighted mean success probability over non-physical and physical tasks: row 1 = $w_N \bar{P}_N$; row 2 = $\bar{P}_N$; row 3 = $w_N \bar{P}_N + (1 - w_N) \bar{P}_P$. SOC 15 rows provide an in-domain benchmark.

Measure (share of wage bill in...)	Denominator	Frontier model	
		Claude Opus 4.6 (Feb 2026; p50 12.0 h, p80 1.2 h)	Mythos Preview (early) (Apr 2026; p50 17.4 h, p80 3.1 h)
Expected success	all tasks	52%	60%
Expected success	non-physical tasks	67%	77%
Expected success, physical tasks scored by the curve rather than excluded	all tasks	70%	79%
Shorter than 80% horizon	all tasks	9%	31%
Shorter than 80% horizon	non-physical tasks	11%	40%
Shorter than 50% horizon	all tasks	75%	78%
Shorter than 50% horizon	non-physical tasks	96%	100%
Observed in Claude usage, May 2026	all tasks	25%	25%
Observed in any AEI release, Dec 2024 to May 2026	all tasks	44%	44%
Expected success, in-domain check	SOC 15 tasks	60%	69%
Shorter than 80% horizon, in-domain check	SOC 15 tasks	3%	17%
Observed in any AEI release, in-domain check	SOC 15 tasks	76%	76%

4.1 The task duration distribution

Figure 3 shows why the threshold shares behave so differently. Most of the wage bill falls between 15 minutes and 16 hours. Because the 50 per cent horizons of both models sit to the right of this mass, the p_{50} threshold captures nearly all non-physical tasks.

This near-saturation reflects the AEI duration data, not a ceiling on capability. AEI durations measure modular pieces of work delegated in chat, which are short by selection. Being shorter

Table 4: Compound 95 per cent intervals from 500 nested bootstrap draws, accounting for METR curve fit uncertainty, AEI duration measurement noise and imputation variance. These are sampling intervals; they do not reflect the choice of success concept or of the excluded set.

Measure	Error source resampled	Claude Opus 4.6 point (95% interval)	Mythos Preview (early) point (95% interval)
Expected success, all tasks	METR curve (family, task, run)	52% (45 to 60)	60% (53 to 67)
	Durations (measurement noise and imputation)	52% (51 to 52)	60% (59 to 60)
	Both	52% (44 to 60)	60% (52 to 66)
Shorter than 80% horizon, all tasks	METR curve (family, task, run)	9% (1 to 25)	31% (13 to 60)
	Durations (measurement noise and imputation)	9% (7 to 11)	31% (28 to 34)
	Both	9% (1 to 26)	31% (14 to 59)
Expected success, non-physical tasks	Both	67% (56 to 76)	77% (66 to 85)

than the p_{50} horizon only means that the fitted curve gives better-than-even odds; these same non-physical tasks have a wage-weighted expected success of 67 per cent under Claude Opus 4.6 and 77 per cent under Mythos Preview (early), not 100 per cent.

Because METR flags horizons beyond 16 hours as unreliable, the integrated expected-success share is the more stable aggregate. The p_{80} share is sensitive to the model’s confidence interval because its horizon crosses the densest region of the duration distribution (Appendix E).

4.2 Capability over time

Figure 4 traces the capability measures as the frontier advanced from 2023 to 2026. The expected-success share grew from 17 per cent under o1 to 60 per cent under Mythos Preview (early). The discrete jump in the p_{80} threshold share at the last step has three sources, set out in Appendix E.

Because the expected-success share depends on the curve only through its 50 per cent horizon and its slope, it can be read off for any future METR estimate (Appendix Figure 12). At the median slope, the share reaches 70 per cent at a horizon of about 56 hours, roughly 7 months beyond Mythos Preview (early) at the current doubling rate, and then approaches its ceiling of 78 per cent, the non-physical share of the wage bill, slowly.

4.3 Capability and observed use

Figure 5(a) compares capability with use across each AEI data window. Observed use within any single window plateaued between 21 and 27 per cent of the wage bill, while the cumulative union across releases has grown to 44 per cent. The cumulative measure depends on the definition of use: requiring a task to reach one in ten thousand conversations cuts the union to 20 per cent, while adding API traffic raises it to 49 per cent (Appendix Table 15).

Figure 5(b) cross-tabulates the May 2026 wage bill. Observed use is not a subset of capability. At the 80 per cent horizon of Mythos Preview (early), 26 per cent of the wage bill sits beyond the capability frontier but still appears in Claude usage, which indicates that workers use AI on pieces of tasks the agent cannot complete alone. 16 per cent of the wage bill falls inside the frontier but

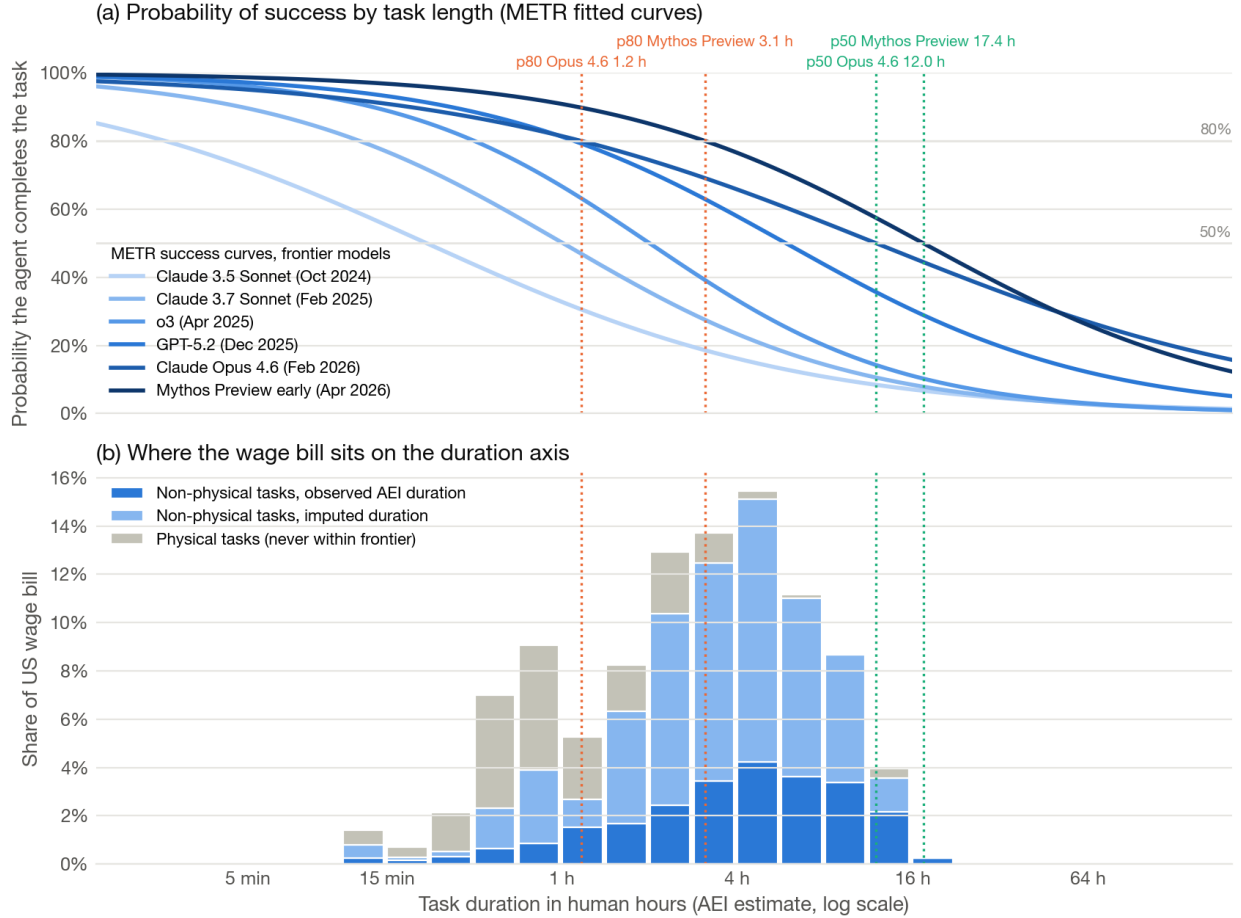


Figure 3: (a) METR’s fitted success curves for six frontier models. (b) The US wage bill by task duration on the same axis, in half-log₂ bins. Dotted lines: the 50 and 80 per cent horizons. The two 80 per cent horizons bracket 22 per cent of the wage bill.

is unused.

The gap between capability and use is therefore not a simple adoption lag. The usage measure includes augmentation beyond the frontier and excludes unused capability within it. AEI data are also extensive-margin only: they record whether a task appears in conversations, not the intensity or the labour hours of the AI’s contribution.

4.4 Heterogeneity by occupation group

Figure 6 shows these measures across the 22 SOC major groups. Sales, legal, office, business and computer occupations have the highest expected-success shares and the highest observed use. Construction, cleaning, food service and installation occupations rank lowest, because of the physical filter. Within the in-domain Computer and Mathematical occupations, the expected-success share is 69 per cent, and 76 per cent of the group’s wage bill appears in cumulative Claude usage.

Appendix F varies the assumptions one at a time. Across the variants that keep the baseline denominator, and leaving aside the in-person rules of Section 2.4, the expected-success share ranges from 49 to 66 per cent, with the exclusion rule and the duration scaling producing the largest shifts.

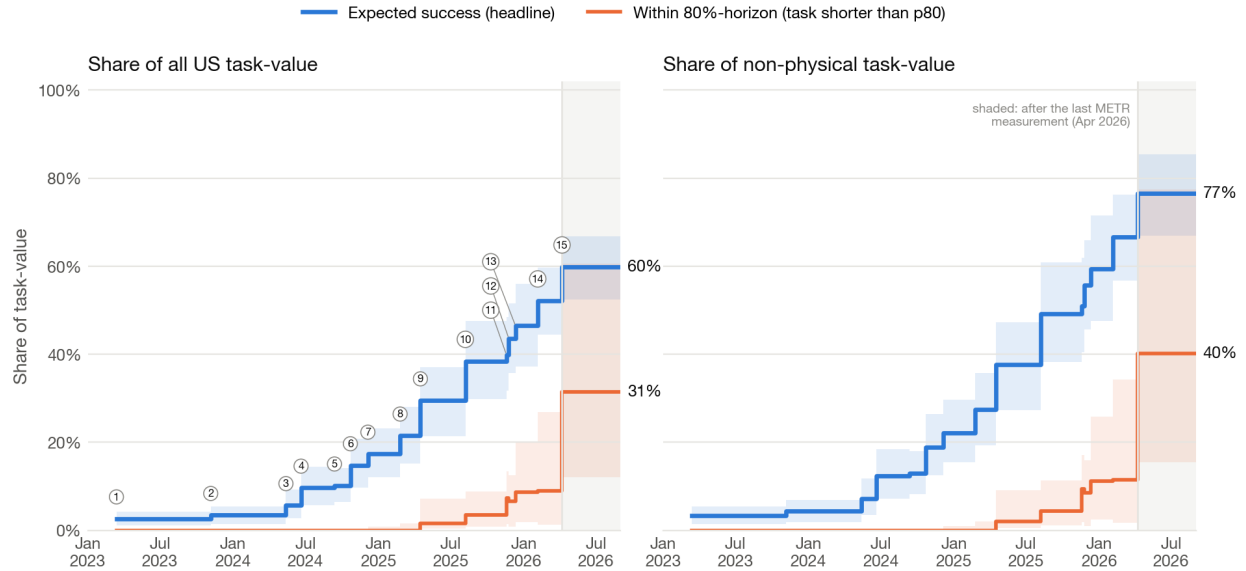


Figure 4: Share of the wage bill in tasks the frontier agent is expected to complete (blue) and in tasks shorter than the 80 per cent horizon (orange). Left: denominator is all tasks. Right: denominator is non-physical tasks only. Frontier = longest 50 per cent horizon among models released by each date. Bands = 95 per cent intervals from the compound bootstrap.

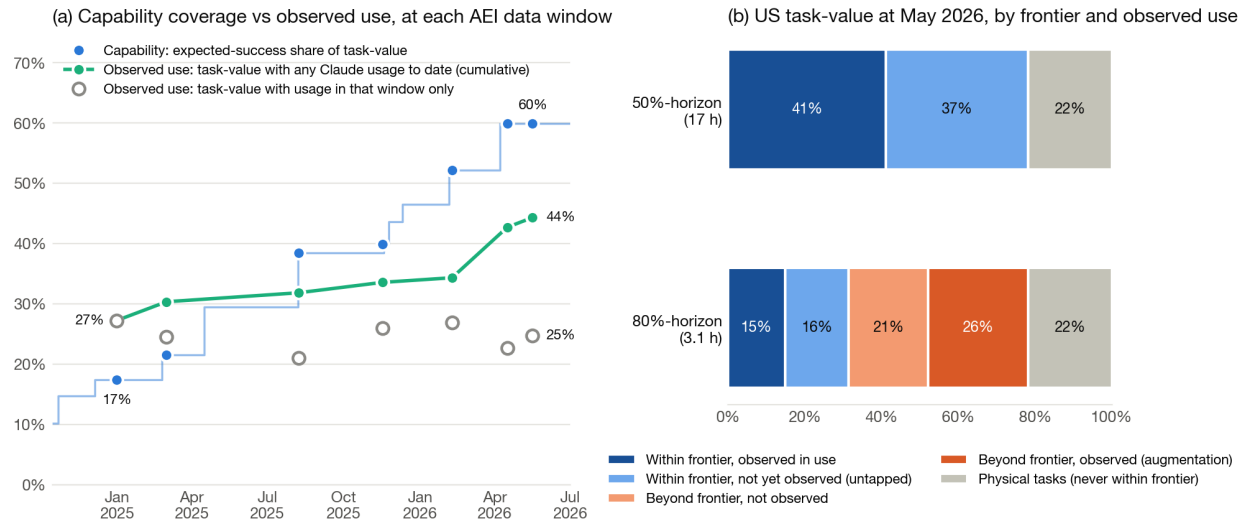


Figure 5: (a) At each AEI data window: expected-success share of the wage bill under the frontier as of the window's last day (blue); share of the wage bill in tasks with a published Claude.ai usage share in any window to date (green) and in that window only (hollow grey). (b) The May 2026 wage bill split by whether a task is shorter than the horizon and whether it has appeared in any AEI release, for the 50 and 80 per cent horizons of Mythos Preview (early).

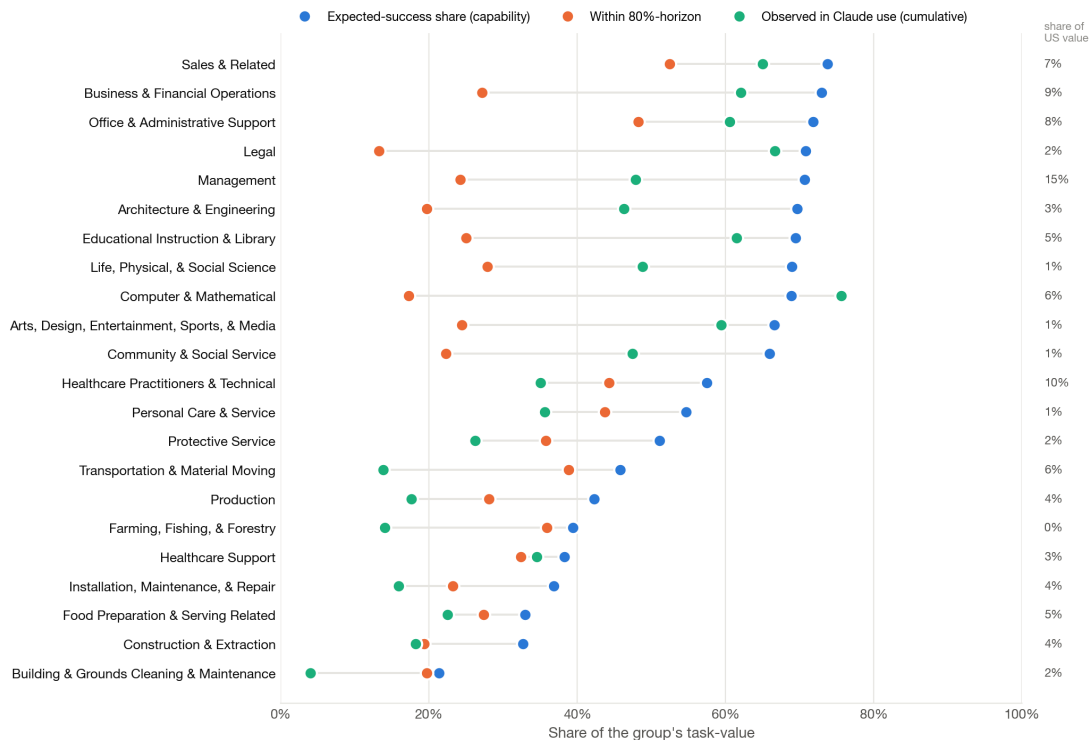


Figure 6: By SOC major group, under Mythos Preview (early): expected-success share (blue), share in tasks shorter than the 80 per cent horizon (orange), share with a published usage share in any AEI release (green). Denominator: the group’s own wage bill, physical tasks included.

5 Discussion

5.1 From tasks to jobs

An occupation is a bundle of tasks, so coverage at the level of tasks does not translate directly into occupational exposure. For each 6-digit occupation I compute its own expected-success share and report the share of US employment above given thresholds (Table 16). Here I count only occupations that can be done remotely and without close physical contact (Section 2.4): the baseline filter’s blind spot for setting falls on exactly the occupations that would otherwise top this ranking (Appendix Table 21). Under Mythos Preview (early), 29 per cent of US employment is in occupations where at least half of task value is expected to succeed, and 8 per cent where at least three quarters is. The 66 per cent of employment in occupations that cannot be done remotely or need close contact counts as zero. On the baseline filter the first figure would be 64 per cent. Figure 7 shows the same distribution for a stricter measure, the share of task value in tasks shorter than the 80 per cent horizon.

No occupation reaches 90 per cent of its task value under either model; the highest share is 85 per cent. The ceiling comes from the physical tasks that nearly every occupation retains, not from the success curve: on the non-physical denominator the highest share is 91 per cent. Table 5 lists the occupations with the highest shares, which are clerical and dispatch occupations whose tasks are short. Their durations are mostly imputed from similar work activities, so the order within the top of the table is not precise.

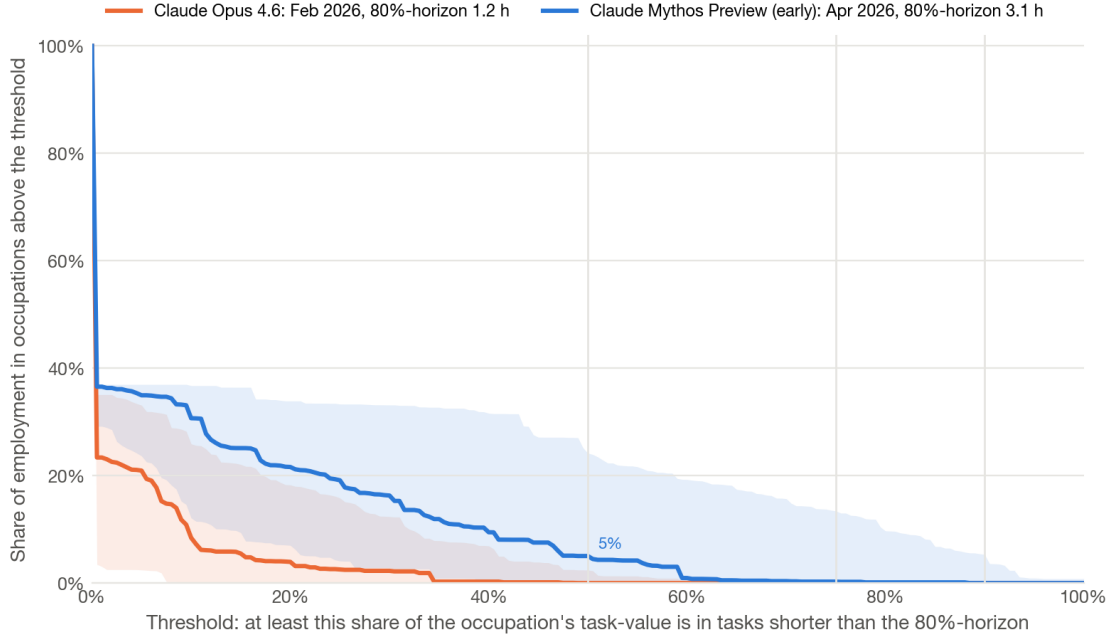


Figure 7: Share of employment in 6-digit occupations where at least the threshold share of the occupation’s task value is in tasks shorter than the frontier model’s 80 per cent horizon. Physical tasks, and every task of an occupation that cannot be done remotely (Dingel and Neiman, 2020) or needs close physical contact, count against the occupation. Denominator: 153.4 million jobs. Bands: 95 per cent bootstrap intervals from the frontier model’s fitted curve.

The additive aggregate measures displaced labour accurately only if tasks can be unbundled and removed from a job one at a time. If the uncovered tasks are the core of the job, high coverage at the task level may change occupational demand very little until the whole bundle is covered (Autor, 2015); whether it raises or lowers the value of the remaining work depends on whether the covered tasks are the expert or the inexperienced ones (Autor and Thompson, 2025), and linear aggregation overstates displacement when tasks are complements (Gans and Goldfarb, 2026). With a weakest-link aggregator (the lowest success probability among an occupation’s more important tasks), only 20 per cent of employment is above the 50 per cent threshold.

An expected-success share of 60 per cent therefore means partial capability across the occupations that can be done remotely, rather than the complete displacement of a few. Whether partial coverage turns into substitution or augmentation requires data on how firms reorganise work, which are beyond the scope of this paper.

5.2 Comparison with exposure measures

On the baseline filter, across 782 occupations, the rank correlation between the expected-success share and the value-weighted Eloundou et al. (2024) exposure score is 0.79. Most of this agreement comes from the physical filter; conditional on being non-physical, the rank correlation is -0.48. The duration axis demotes occupations whose non-physical tasks are long, such as software developers (median task duration 8.1 hours), and promotes clerical and front-line service occupations whose tasks are short. The long durations of software work partly reflect what users delegate: durations of coding-heavy tasks rose most when the AEI data began to include Cowork (Appendix K). Figure 11

Table 5: The ten occupations with the highest expected-success share of task value under Mythos Preview (early), and the five largest occupations by employment. Physical tasks, and every task of an occupation that cannot be done remotely (Dingel and Neiman, 2020) or needs close physical contact, count as zero; Appendix Table 21 gives the same table on the baseline filter. Weakest link: lowest success probability among the occupation’s above-median-importance tasks, zero if any is excluded. Median duration: wage-weighted baseline durations.

SOC	Occupation	Employment (thousands)	Expected success	Weakest link	Observed use	Median duration (h)
43-3011	Bill and Account Collectors	159	85%	72%	63%	1.7
43-4151	Order Clerks	75	85%	74%	45%	1.7
43-4021	Correspondence Clerks	4	85%	74%	72%	1.7
43-5032	Dispatchers, Except Police, Fire, and Ambulance	203	84%	73%	39%	2.1
41-9041	Telemarketers	58	84%	80%	75%	2.7
29-2072	Medical Records Specialists	195	82%	0%	65%	1.5
43-4011	Brokerage Clerks	36	82%	73%	100%	2.0
43-3061	Procurement Clerks	56	81%	72%	62%	2.8
43-9041	Insurance Claims and Policy Processing Clerks	214	81%	69%	54%	3.4
39-1022	First-Line Supervisors of Personal Service Workers	114	81%	62%	31%	2.6
31-1120	Home Health and Personal Care Aides	4,306	0%	0%	50%	1.5
41-2031	Retail Salespersons	3,898	0%	0%	58%	2.1
35-3023	Fast Food and Counter Workers	3,854	0%	0%	16%	0.6
11-1021	General and Operations Managers	3,503	69%	62%	34%	7.0
29-1141	Registered Nurses	3,380	14%	0%	39%	2.9

in Appendix G plots each of these relationships.

5.3 Relation to scenario models

Korinek et al. (2026) define the affected mass m_t as the share of the economy’s tasks, at pre-AI wage-bill weights, that AI affects, all of them in cognitive occupations; the automation share ψ_t is the fraction of AI-performed work that is automated rather than augmented. Table 6 sets this paper’s measures beside their values. Their mid-2026 anchor of 0.14 is an observed-exposure measure built from Claude usage (Massenkoff and McCrory, 2026); I reproduce it from the AEI files (0.22 in cognitive occupations, 0.01 elsewhere). My capability measure is much larger. Restricted to their cognitive occupations, the expected-success share under Mythos Preview (early) is 0.48, close to their extreme 2030 value of 0.5, and 0.31 for cognitive occupations that can be done remotely and without close contact. The ceilings are of similar size: all cognitive work is 0.624 of tasks in their calibration, and all non-physical work is 78 per cent of the wage bill in mine.

On METR’s definition of success, then, capability is not what separates their scenarios; the diffusion share, the productivity gain per task and the automation share do. For the automation share, AEI classifies about half of Claude.ai use as automation (0.49), close to their modest value, but 0.94 of API use, above their extreme value of 0.90. A capability reading of ψ_t , the share of expected success in tasks the curve completes at least four times in five, is 0.45 under Mythos Preview (early) and 0.15 under Claude Opus 4.6. Under a stricter standard, such as a client’s acceptance of a whole project, the capability reading of m_t would be much lower (Section 5.4).

Table 6: This paper’s measures beside the scenario parameters of Korinek et al. (2026). Their values are published figures (anchor, 2030 scenarios, ceiling and automation shares, their Table 1). Affected mass is a share of the wage bill; “cognitive occupations” are their twelve SOC major groups. Expected success uses METR’s success concept and Mythos Preview (early). The AEI automation share is the usage-weighted share of conversations classified as automation (directive or feedback-loop use) in the June 2026 release.

	Korinek et al. (2026)	This paper
<i>Affected mass m (share of the wage bill)</i>		
Anchor: their observed exposure; our Claude.ai use, May 2026	0.14	0.25
Mid-2026 expected success, cognitive occupations		0.48
of which in remote jobs without close contact		0.31
Mid-2026 expected success, all occupations		0.60
2030 scenarios: modest / substantial / extreme	0.2 / 0.3 / 0.5	
Ceiling: all cognitive work / all non-physical work	0.624	0.78
<i>Automation share ψ (AI-performed work automated rather than augmented)</i>		
Scenarios: modest / substantial / extreme	0.50 / 0.75 / 0.90	
AEI automation share of use, June 2026: Claude.ai / API		0.49 / 0.94
Share of expected success with $P \geq 0.8$: Opus 4.6 / Mythos		0.15 / 0.45

5.4 Limitations

Four limitations matter most. The first is the duration mismatch. METR measures human time for complete, well-specified tasks performed by contractors, whereas AEI estimates time for the pieces of work users delegate in conversations. Delegated pieces are short by selection, so AEI durations may understate the length of employer-defined tasks and bias the capability shares upward. Doubling every duration lowers the expected-success share under Mythos Preview (early) to 52 per cent (Appendix F). Claude’s time estimates correlate with actual task times on software tickets about as well as developers’ own estimates (Tamkin and McCrory, 2025), but the AEI durations also depend on the platform and on which products the data cover (Appendix K). Only 26 per cent of the wage bill has an observed duration; hold-out checks show the imputation is unbiased within the observed sample (Appendix Table 14), but not whether it transfers to tasks that are never delegated.

Second, the success curve is extrapolated beyond software. Within the suite, the curve overstates success in the one-to-four-hour band; reweighting the residuals suggests this alone inflates the headline share by about 2 points (Appendix Table 9). Outside the suite, AEI’s own success data do not show the curve’s decline with task length (Section 3), and the top of the frontier remains poorly measured, which makes the 80 per cent threshold share uncertain (Appendix E).

Third, the level depends on what counts as success. METR scores well-specified tasks automatically. Where a client must accept a whole freelance project, METR’s curves predict completion of 30 to 51 per cent of projects against observed rates of 1.2 to 4.2 per cent (Mazeika et al., 2025). Where a user works with Claude in conversation and can iterate, success falls more slowly with task length than METR’s curves imply (Section 3). Neither standard supplies a success curve for the tasks in the economy: the Remote Labor Index gives one rate per model over 240 projects, and the AEI labels cover only the work users chose to delegate, for the models that served it. Appendix H reads each gap as a shift in METR’s horizon. The implied shares vary widely and rest on that reading, so I treat them as illustrations rather than alternative estimates. Even so, the gaps between standards are far larger than the sampling interval in Table 4.

Fourth, the measure gives a task the agent completes with probability 0.7 credit for 0.7 of its value,

with no allowance for the time a worker spends checking the output. If checking takes a quarter of the task’s human time, the net share falls to 40 per cent (Appendix L), and only 8.8 per cent of the wage bill is in tasks the curve expects to succeed nine times in ten. Experienced developers in METR’s own trial were slower with AI than without it (Becker et al., 2025).

In expectation, then, the capability frontier on METR’s definition covers a majority of the wage bill, while observed use covers less than half. The coverage is broad but partial: the frontier reaches parts of most jobs that can be done remotely and the whole of few. Refining the estimates requires human completion times for tasks defined at the level of the occupation, and success curves measured outside software under a stated standard of success.

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A Data Construction

The estimates in this paper are constructed from a single task-level dataset. Every aggregate reported in the text derives from an analysis file with one row per O*NET task, which records the task’s wage weight, observed or imputed AEI duration, physical classification, Eloundou et al. (2024) exposure label, usage flags and fitted success probabilities under the frontier models.

- **METR time horizons:** Data are downloaded from metr.org/time-horizons (benchmark_results_1_1.yaml and task_results_1_1.yaml, updated 8 May 2026). The curve refits follow METR’s public methodology (github.com/METR/eval-analysis-public): a weighted logistic regression of run success on $\log_2$ human minutes, with weights equal to the family weight divided by runs per task, normalised to one per model, using 10^{-5} regularisation. Standard errors and intervals are generated via 500 bootstrap draws over family, task, and run. The frontier at date d is defined as the model with the longest 50 per cent horizon released on or before d , and an AEI data window is paired with the frontier as of its last day. For the robust variants (Appendix F), the curves are refit using a probit link, logistic functions with asymptotes fixed at 0.85, 0.90, or 0.95, HCAST tasks only, and a leave-one-family-out procedure.
- **Anthropic Economic Index (AEI):** Sourced from huggingface.co/datasets/Anthropic/EconomicIndex. Usage shares are extracted from seven Claude.ai data windows spanning December 2024 to May 2026. Per-task human time (human_only_time_mean, which averages the

estimated hours for a human to complete the delegated work without AI) uses the June 2026 release as the baseline, as it uniquely provides exact O*NET task IDs. Releases prior to June 2026 identify tasks by text string; I recover 87 to 91 per cent of usage via exact string matching and drop the residual (mostly retired O*NET occupations). Missing durations are imputed using the simple mean over a task’s detailed work activities of the geometric mean of observed durations in each activity (36 per cent of the wage bill). If no activity has an observed peer, the imputation backs off sequentially to the intermediate (22 per cent) and generalized work activity (15 per cent), and finally to the global geometric mean (0.8 per cent). Tasks map to 1.27 detailed work activities on average.

- **O*NET 30.2:** Task statements, ratings (importance, relevance, frequency), and the work-activity hierarchy are merged to construct the wage weights. These weights use O*NET’s core tasks (relevance ≥ 67 per cent and mean importance ≥ 3.0) with positive relevance; specialties are averaged into the 6-digit SOC code, and 29 occupations missing ratings receive equal weights. Physical tasks are defined by mapping to the following generalized work activities: 4.A.3.a.1–4 (performing general physical activities, handling and moving objects, controlling machines and processes, operating vehicles and equipment), 4.A.3.b.4–5 (repairing and maintaining equipment) and 4.A.4.a.5 (assisting and caring for others). The Dingel–Neiman rule uses the Work Context and Work Activities files of the same release (Appendix J).
- **OEWS May 2025 (National):** Provides employment and mean annual wage by detailed occupation. Hourly wages are annualised by OEWS at 2,080 hours. Twelve OEWS hybrid codes are mapped explicitly. Occupations lacking O*NET task data (accounting for 1.3 per cent of employment) are dropped.
- **Eloundou et al. (2024) labels:** Sourced from github.com/openai/GPTs-are-GPTs and used solely to define the alternative non-physical set and for the rank correlation comparison in Section 5.2.
- **External benchmarks:** METR’s cross-domain horizons are taken from github.com/METR/cross-domain-horizon (pinned commit). Published figures from the Remote Labor Index (Mazeika et al., 2025), GDPval (Patwardhan et al., 2025), the AEI January 2026 report (Appel et al., 2026) and Korinek et al. (2026) are recorded, with page references and access dates, in the replication files; they enter only Appendix H and Section 5.3.

An interactive version of the task-level data, in which all 18,796 tasks are drawn as individual marks arranged by occupation group and duration against the frontier, is available at <https://aitasks-ecru.vercel.app/>.

B METR Task Suite and Refit Verification

Table 7 gives the composition of METR’s Time Horizon 1.1 suite by source. Table 8 reports METR’s published 50 and 80 per cent horizons alongside the refit, and the fitted slope per doubling of task length, for all models with fitted curves. “Frontier at release” is METR’s `is_sota` indicator; these models are the frontier steps in Figure 4. Table 9 reports the fit’s residuals by duration bin.

Table 7: Composition of METR’s Time Horizon 1.1 task suite. Weight = share of METR’s diversity-adjusted weight in the Mythos Preview (early) fit, where each task in a family of n_f tasks carries weight $1/\sqrt{n_f}$, so a family’s total weight is proportional to $\sqrt{n_f}$. The band from half an hour to eight hours holds 40 per cent of the weight on 72 tasks.

Source	Content	Tasks	Families	Human time: median, max	Weight
HCAST	software engineering, machine learning, cybersecurity, and computer-based research tasks	157	69	44 min, 30.0 h	81%
RE-Bench	machine-learning research engineering tasks	5	5	8.0 h, 8.2 h	4%
SWAA	seconds-long software actions (code completion, file selection, request routing)	66	5	5 s, 43 s	15%

Table 8: METR’s published 50 and 80 per cent horizons and my refit from the per-task results, all models with fitted curves. Slope = the refit coefficient on $\log_2$ human minutes, sign reversed: the fall in the log-odds of success per doubling of task length. “Frontier at release” is METR’s `is_sota` flag; these models are the numbered steps in Figure 4, in date order from GPT-4 (March 2023). Mythos Preview (early) is a pre-release checkpoint measured on 1,149 runs.

Model	Released	p50 METR (min)	p50 refit (min)	p80 METR (min)	p80 refit (min)	Slope	Frontier at release
GPT-4 0314	2023-03-14	4.0	4.0	0.9	0.9	0.64	yes
GPT-4 1106	2023-11-06	4.0	4.0	0.8	0.8	0.59	yes
Claude 3 Opus	2024-03-04	4.0	4.0	0.6	0.6	0.53	
GPT-4 Turbo	2024-04-09	3.7	3.7	0.9	0.9	0.69	
GPT-4o	2024-05-13	7.0	7.0	1.3	1.3	0.56	yes
Claude 3.5 Sonnet (Old)	2024-06-20	11.4	11.4	1.7	1.7	0.50	yes
o1-preview	2024-09-12	20.3	20.3	4.4	4.4	0.63	yes
Claude 3.5 Sonnet (New)	2024-10-22	20.5	20.5	2.6	2.6	0.46	yes
o1	2024-12-05	38.8	38.8	7.1	7.1	0.57	yes
Claude 3.7 Sonnet	2025-02-24	60.4	60.4	12.1	12.1	0.60	yes
o3	2025-04-16	119.7	119.7	30.0	30.0	0.69	yes
Claude 4 Opus	2025-05-22	100.4	100.4	20.4	20.4	0.60	
Claude 4.1 Opus	2025-08-05	100.5	100.5	23.5	23.5	0.66	
GPT-5	2025-08-07	203.0	203.0	38.3	38.3	0.58	yes
Gemini 3 Pro	2025-11-18	224.3	224.3	54.1	54.1	0.68	yes
GPT-5.1-Codex-Max	2025-11-19	223.7	223.7	50.6	50.6	0.65	
Claude Opus 4.5	2025-11-24	293.0	293.0	49.4	49.3	0.54	yes
GPT-5.2	2025-12-11	352.2	352.3	66.0	66.0	0.57	yes
GPT-5.3-Codex	2026-02-05	349.5	349.5	54.7	54.7	0.52	
Claude Opus 4.6	2026-02-05	718.8	718.8	69.9	69.9	0.41	yes
Gemini 3.1 Pro	2026-02-19	384.1	384.1	89.8	89.8	0.66	
GPT-5.4	2026-03-05	341.7	341.7	53.9	53.9	0.52	
Claude Mythos Preview (early)	2026-04-07	1044.8	1044.7	185.9	185.9	0.56	yes

C Task Examples and Classification

Table 10 lists the twelve METR task families with the largest weight in the fit; the family names are METR’s identifiers, and Rein et al. (2025) and Wijk et al. (2024) describe the tasks. Table 11 shows, for eight SOC major groups, the O*NET task with the largest wage bill in the group, with its duration on the baseline axis. Figure 8 shows every O*NET task of one occupation, Registered Nurses, with the physical filter applied.



Figure 8: The 27 O*NET tasks of Registered Nurses (O*NET-SOC 29-1141.00; nursing specialties excluded here). Each tile is one task statement, shortened. Dark blue: no physical work activity. Grey with orange border: some physical work activities, so counted as physical. Grey: every work activity physical. Physical tasks never count as within the frontier. Grey text: the task's AEI human duration on the baseline axis, the estimated human time for the work delegated in a conversation, averaged over conversations; * = imputed. Tiles are ordered by classification and then by O*NET importance.

Table 9: Observed minus fitted success by duration bin within METR’s suite, Mythos Preview (early). Observed and fitted are diversity-weighted means over the tasks in the bin; difference in percentage points. Last column: share of the total wage bill in non-physical tasks whose AEI duration falls in the bin. Reweighting the differences by the last column gives the change in the expected-success share that would follow if the suite’s residual pattern held in the economy: about -2 points for Mythos Preview (early) and -1 for Claude Opus 4.6.

Human time	METR suite		Mythos Preview (early)			Non-physical wage bill in this bin
	Tasks	Weight	Observed success	Fitted	Difference	
< 1 min	67	15%	99%	100%	-0.5	0%
1-15 min	52	24%	99%	98%	+0.3	1%
15-60 min	36	17%	98%	94%	+3.3	7%
1-4 h	18	9%	76%	86%	-9.9	32%
4-16 h	50	31%	68%	67%	+1.3	38%
> 16 h	5	3%	45%	47%	-1.4	0%

Table 10: The twelve METR task families with the largest diversity-adjusted weight in the Mythos Preview (early) fit.

Task family	Source	Tasks	Human time, shortest to longest	Weight
arithmetic	SWAA	20	1 s to 43 s	3.7%
file_selection	SWAA	20	2 s to 6 s	3.7%
alert_triage	SWAA	10	4 s to 10 s	2.6%
env_scientist	HCAST	9	44 min to 11.1 h	2.5%
request_routing	SWAA	9	8 s to 18 s	2.5%
pico_ctf	HCAST	7	2 min to 1.1 h	2.2%
data_cleaning_arjun	HCAST	7	6 min to 6 min	2.2%
code_completion	SWAA	7	3 s to 6 s	2.2%
palisade_crackme	HCAST	6	21 min to 8.0 h	2.0%
blackbox	HCAST	6	10 min to 1.1 h	2.0%
password_check	HCAST	6	57 s to 32 min	2.0%
hypothesis_testing	HCAST	6	9 min to 49 min	2.0%

D Variation within METR’s Suite

Figure 9 plots, for the two frontier models, each METR task’s observed success rate against its human time, with METR’s pooled curve and curves fitted within each task source. Table 12 gives the horizons from each fit and the range of horizons obtained by dropping any one task family and refitting. The R^2 in the panel titles is the weighted R^2 of task success rates on the pooled curve; the corresponding ceiling if the curve were exactly right, given the number of runs per task, is 0.61 for Claude Opus 4.6 and 0.57 for Mythos Preview (early). Across all 23 models with fitted curves the highest weighted R^2 is 0.68 (GPT-4 0314, ceiling 0.89); the frontier models sit low because most of their tasks are solved nearly every time and the remaining failures are concentrated in particular tasks rather than at particular lengths. The RE-Bench fits are not shown because five tasks do not identify a slope. The SWAA-only horizons are shown but should not be read as measurements: the longest SWAA task takes 43 seconds, so both horizons for Mythos Preview (early) and the 50 per cent horizon for Claude Opus 4.6 lie beyond the subsuite’s data, and one percentage point of observed success separates a one-minute from a sixteen-minute fitted horizon.

Table 11: The O*NET task with the largest wage bill in each of eight SOC major groups. Hours = AEI human duration on the baseline axis: the estimated time for a person to do, without AI, the piece of the task that users delegate in one conversation, averaged over conversations mapped to the task; * = imputed. The wage weight attaches to the whole statement. Physical = at least one of the task’s work activities is physical, so the task never counts as within the frontier.

Occupation	Task (O*NET statement)	Hours	Physical
General and Operations Managers	Review financial statements, sales or activity reports, or other performance data to measure productivity or goal. . .	7.0	
Software Developers	Analyze user needs and software requirements to determine feasibility of design within time and cost constraints.	11.3	
Lawyers	Interpret laws, rulings and regulations for individuals and businesses.	4.3	
Registered Nurses	Record patients’ medical information and vital signs.	2.0*	
Retail Salespersons	Greet customers and ascertain what each customer wants or needs.	0.5	yes
Customer Service Representatives	Confer with customers by telephone or in person to provide information about products or services, take or enter orders. . .	0.8	
Construction Laborers	Tend pumps, compressors, or generators to provide power for tools, machinery, or equipment or to heat or move materials. . .	2.1*	yes
Laborers and Freight, Stock, and Material Movers, Hand	Maintain equipment storage areas to ensure that inventory is protected.	0.6*	

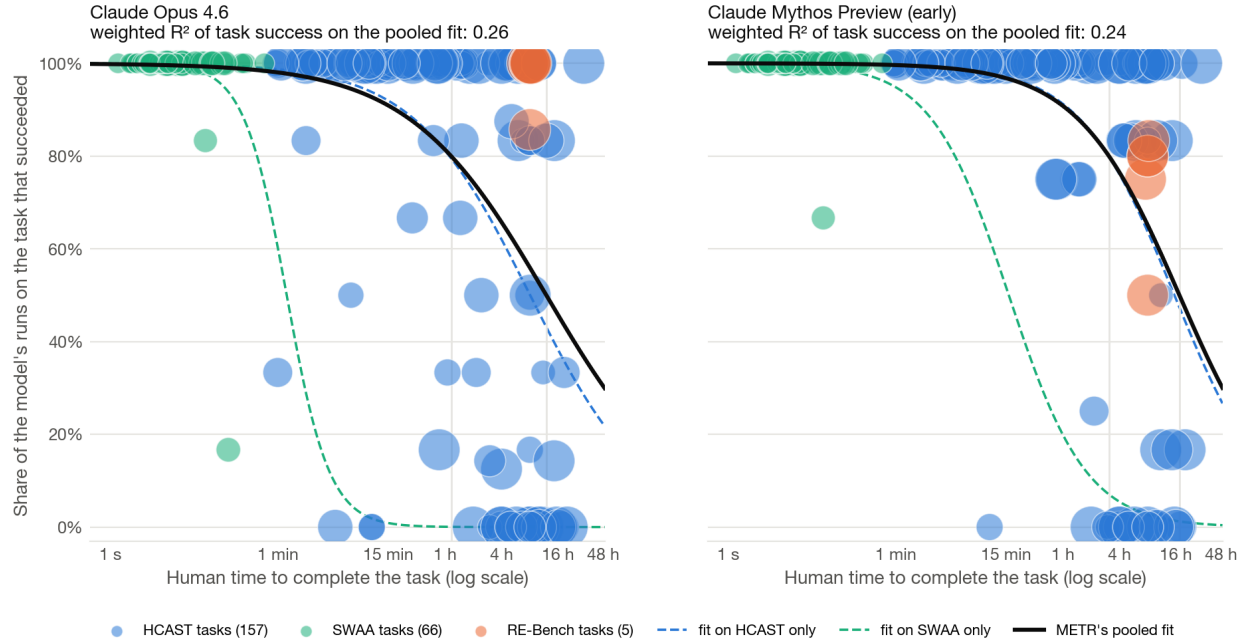


Figure 9: Each point is one METR task; area is proportional to METR’s diversity weight. Black: METR’s pooled fit. Dashed: logistic fits within one task source. Grey lines: the pooled 50 and 80 per cent horizons. Left: Claude Opus 4.6. Right: Mythos Preview (early).

Table 12: Horizons from METR’s pooled fit, from fits within one task source, and the range when any one of the 79 task families is dropped. Success = diversity-weighted share of runs that succeeded. †: horizon lies beyond the longest task in the subsuite (an extrapolation, not a reading within the data).

Model	Fit	Tasks	Weight	Success	50% horizon	80% horizon
Claude Opus 4.6	pooled (METR’s fit)	228	100%	79%	12.0 h	1.2 h
	HCAST tasks only	157	81%	74%	8.1 h	1.1 h
	SWAA tasks only	66	15%	98%	1 min [†]	36 s
	RE-Bench tasks only	5	4%	97%	n/a	n/a
	drop any one family				10.3 h to 14.2 h	1.1 h to 1.4 h
Mythos Preview (early)	pooled (METR’s fit)	228	100%	85%	17.4 h	3.1 h
	HCAST tasks only	157	81%	83%	15.4 h	3.1 h
	SWAA tasks only	66	15%	99%	16 min [†]	4 min [†]
	RE-Bench tasks only	5	4%	74%	n/a	n/a
	drop any one family				14.8 h to 20.9 h	2.9 h to 3.6 h

E Decomposing the April 2026 Capability Step

The 80 per cent threshold share in Figure 4 jumps from 9 to 31 per cent when the frontier passes from Claude Opus 4.6 to Mythos Preview (early). The step is a faithful reading of METR’s published fits, but three things make it as large as it is. First, the frontier only moves when a model beats the best 50 per cent horizon to date. METR measured two other models between these two: Gemini 3.1 Pro (19 February 2026; 50 per cent horizon 6.4 hours, 80 per cent horizon 1.5 hours) and GPT-5.4 (5 March 2026; 50 per cent horizon 5.7 hours, 80 per cent horizon 0.9 hours). Neither beat the 50 per cent horizon of Claude Opus 4.6, so neither appears in the figure, although the 80 per cent horizon of Gemini 3.1 Pro was the longer of the two and would have put 11 per cent of the wage bill below it; its expected-success share, 49 per cent, is below Claude Opus 4.6’s, so the frontier rule and the share agree on the ranking. Second, Claude Opus 4.6 has an unusually flat success curve, 0.41 per doubling against 0.56 for Mythos Preview (early). Its 50 per cent horizon is 10 times its 80 per cent horizon, against 4 to 6 times for the other frontier models since 2025, so its 80 per cent horizon barely moved from that of GPT-5.2 (1.1 to 1.2 hours) while its 50 per cent horizon doubled. Much of the 8-point rise in the expected-success share between the two models is therefore the slope steepening in the half-hour-to-eight-hour band rather than the horizon lengthening. Third, the 80 per cent threshold share counts tasks shorter than a threshold, and the threshold moved from 1.2 to 3.1 hours, across the part of the duration axis where the wage bill is: 22 per cent of the wage bill is in non-physical tasks with durations between those two values (Figure 3(b)), and 32 per cent is in non-physical tasks between one and four hours. The expected-success share integrates the curve instead of counting a threshold, and moves 8 points on the same step.

The threshold is also poorly determined at the top. The 80 per cent horizon of Mythos Preview (early) has a 95 per cent interval of 1.6 to 6.6 hours, which puts the share anywhere between 12 and 61 per cent (the orange band in Figure 4). Under Claude Opus 4.6 the interval is 1 to 27 per cent. The level of the orange line in 2026 is therefore uncertain by tens of points either way, and the size of the step says as much about where the wage bill sits on the duration axis as about the model.

F Robustness and Sensitivity Analysis

Table 13 changes one assumption at a time, with Mythos Preview (early) as the frontier, in five groups. Across the variants that keep the baseline denominator, other than the in-person rules, the expected-success share ranges from 49 to 66 per cent. The duration bridge and the exclusion rule matter most. Doubling every duration gives 52 per cent and cuts the 80 per cent threshold share to 12 per cent; halving them raises that share to 58 per cent. Inflating imputed durations alone, which targets the selection of the observed sample rather than the level of the clock, gives 57, 54 and 51 per cent at factors of 1.5, 2 and 3. Using the earlier AEI releases for durations raises the share by 5 points. On the curve side, refitting on HCAST tasks only, dropping any one family, switching to a probit link or fixing the asymptote at 0.85 to 0.95 all leave the share within two points of the baseline, although the probit and low-asymptote fits move the 80 per cent threshold share to 27 and 23 per cent. Task weights within occupations barely matter, including the hours-implied weight. Switching off O*NET’s core filter and weighting every rated task by importance times relevance moves 16 per cent of the wage bill onto supplemental tasks but gives 59 per cent; restricting that to tasks with importance above 2.5 and relevance above 50 per cent gives 59 per cent. Dropping wages and weighting by employment gives 55 per cent. The lenient physical filter gives 63 per cent, the Eloundou labels 49 per cent, and crediting mixed tasks’ non-physical share at a proportionally shorter duration 61 to 62 per cent. The in-person rules of Section 2.4 give 56 per cent at the task level and 35 per cent with the Dingel–Neiman bound. The two observed-use variants leave the expected-success columns unchanged by construction; what they move is the use column, from 44 to 20 per cent under a share threshold and to 49 per cent with API traffic. The last two rows restrict the universe to tasks with an observed duration and are not commensurable with the rest; they are the source of the 75 per cent that appears if the whole column is read as one range.

G Occupation-Level Exposure and Time Series

Figure 10 plots every 6-digit occupation that can be done remotely and without close contact by its expected-success share of task value under Mythos Preview (early) and the share of its task value that has appeared in Claude usage. The 63 per cent of employment in other occupations has an expected-success share of zero and is not shown. Office, business, sales and computer occupations cluster at the right, with high capability and high use; large occupations with high capability but use well below the economy-wide share, such as general and operations managers, make up much of the “inside the horizon, not yet observed” cell of Figure 5(b). Table 16 gives the employment shares behind Section 5.1, and Table 17 the capability and use measures at each AEI data window.

Figure 11 plots the correlations of Section 5.2, on the baseline filter. The agreement in panels (a) to (c) is largely the physical filter, which both measures respect; panels (d) and (e) condition it away by using non-physical value only, leaving a measure that depends on task duration alone. Panel (f) supplies the channel: the occupations the rubric scores as most exposed are the ones whose delegated tasks are longest, and the success curve falls with length. The negative correlation in panel (d) is therefore not disagreement about which occupations are exposed, but the duration axis discounting long analytical work relative to short clerical and service work. Two cautions: the vertical range in panels (d) and (e) is narrow, so the rank correlation turns on small differences and the employment-weighted correlation is much the weaker of the two; and the occupations at the top left of panel (e) are mostly physical occupations whose small non-physical remainder happens

Table 13: One assumption changed at a time; frontier = Mythos Preview (early). Denominator: all tasks, except in the last two rows, where it is the observed-duration tasks. Curve variants refit the same model's runs another way (Appendix A). θ : share of a mixed task's value credited as non-physical, evaluated at duration θD_t . The in-person rows are described in Appendix J.

Variant (one change from baseline)	Expected success	< p80	Observed, cumulative
<i>Duration bridge</i>			
Baseline	60%	31%	44%
Durations from the Nov 2025 and Feb 2026 AEI releases	64%	51%	44%
All durations halved	66%	58%	44%
All durations doubled	52%	12%	44%
Imputed durations only $\times 1.5$	57%	21%	44%
Imputed durations only $\times 2$	54%	17%	44%
Imputed durations only $\times 3$	51%	14%	44%
<i>Success curve (same model, refit)</i>			
Curve fitted on HCAST tasks only	59%	30%	44%
Drop-one-family refit, lowest share (drops <code>mlab</code>)	59%	29%	44%
Drop-one-family refit, highest share (drops <code>spn_cryptanalysis</code>)	61%	37%	44%
Probit link instead of logistic	59%	27%	44%
Logistic with asymptote $\bar{p} = 0.95$ (refit)	60%	35%	44%
Asymptote $\bar{p} = 0.90$	60%	32%	44%
Asymptote $\bar{p} = 0.85$	59%	23%	44%
<i>Task weights and value</i>			
Equal task weights within occupation	60%	31%	44%
Importance $\times$ relevance weights	60%	32%	45%
Importance $\times$ relevance weights, supplemental tasks admitted (no core filter)	59%	30%	43%
Importance $\times$ relevance weights, tasks with importance > 2.5 and relevance $> 50\%$	59%	31%	43%
Frequency-implied weights	60%	32%	46%
Hours-implied weights (duration $\times$ annual frequency)	60%	21%	46%
Employment weights, no wages	55%	33%	39%
<i>Excluded (physical) set</i>			
Non-physical set = Eloundou et al. exposure label > 0	49%	24%	44%
Physical = every work activity physical (lenient filter)	63%	34%	44%
Mixed tasks split: share $\theta = 0.25$ of value at duration θD	61%	32%	44%
Mixed tasks split: $\theta = 0.5$	61%	33%	44%
Mixed tasks split: $\theta = 0.75$	62%	34%	44%
Physical also if any work activity is public-facing or inspecting (task level)	56%	28%	44%
Non-physical set = occupations Dingel and Neiman class as teleworkable	38%	16%	44%
Baseline and Dingel–Neiman teleworkable occupation (occupation-level rule)	35%	14%	44%
and not close contact (O*NET physical proximity < 4)	32%	13%	44%
and not close contact (physical proximity < 4.5)	34%	13%	44%
<i>Observed-use definition (expected success unchanged by construction)</i>			
Observed only if share $\geq 0.01\%$	60%	31%	20%
Observed in Claude.ai or API	60%	31%	49%
<i>Change of universe (different denominator; not part of the one-change range)</i>			
Observed-duration tasks only (denominator changes)	73%	34%	100%
Observed-duration tasks only, Nov 2025 and Feb 2026 AEI releases	75%	56%	100%

Table 14: Leave-one-out check of the duration imputation. Each observed task is re-imputed from the other observed tasks by the baseline rule, at the level the rule would use for it. $\hat{D}_t/D_t$ is the ratio of imputed to observed duration. Value shares are of the wage bill in observed-duration tasks.

Imputation level	Tasks	Share of observed value	Median $\hat{D}_t/D_t$	Mean log ratio, value-weighted	SD of log ratio	Within factor 2
Detailed work activity	2,736	97%	1.00	+0.02	0.49	87%
Intermediate work activity	69	2%	1.57	+0.94	0.78	57%
Generalized work activity	9	0%	2.61	+1.38	0.84	44%
All observed tasks	2,814	100%	1.00	+0.04	0.50	86%

Table 15: Observed use under alternative definitions. Share of the wage bill in tasks meeting the definition; denominator all tasks. Top block: threshold on the task’s share of Claude.ai conversations in a window. Bottom block: number of the seven Claude.ai windows in which the task was published at all.

Definition of “observed”	Tasks	May 2026 window	Any window, Dec 2024 to May 2026
Published at all (baseline)	6,254	25%	44%
Share of conversations $\geq 0.01\%$	2,297	13%	20%
Share $\geq 0.05\%$	700	4%	8%
Share $\geq 0.1\%$	370	2%	4%
Published in at least 1 of the 7 windows	6,254		44%
Published in at least 2 of the 7 windows	5,286		38%
Published in at least 4 of the 7 windows	3,323		24%
Published in at least 7 of the 7 windows	843		9%

to be short.

H External Benchmarks and the Horizon Map

The expected-success share depends on the success curve only through its 50 per cent horizon h and slope b , since $P(D) = \sigma(b \log_2(D/h))$ with $b < 0$. Figure 12 maps any (h, b) to the share; every METR model falls on the surface at its own slope. Holding the slope fixed, dividing the horizon by a factor k is equivalent to multiplying every task duration by k , and at METR’s doubling time it corresponds to $\log_2 k$ doubling periods of frontier progress.

Each external benchmark sets its own standard of success. For each, I find the factor k at which METR’s curve for the same model reproduces the benchmark’s observed success rate at the benchmark’s task lengths, and apply the same factor to the frontier curves (Table 18). The Remote Labor Index publishes the mean (28.9 hours) and median (11.5 hours) completion time of its 240 projects, and I treat project durations as log-normal with those two moments. Table 19 gives the five leaderboard entries that match a METR curve. GDPval tasks take experts 7 hours on average (Patwardhan et al., 2025); its win-or-tie rate against experts is a more generous standard than METR’s, and implies a slightly longer horizon. The AEI usage horizons come from Appel et al. (2026), who find that Claude succeeds on half of API tasks of about 3.5 hours and half of Claude.ai tasks of about 19 hours, against METR’s 2 hours for the same model. METR’s cross-domain horizons use METR’s own estimates by benchmark (Kwa and Cheng, 2025); several assume a slope of 0.6.

These translations are illustrations, not alternative estimates. Each assumes that the whole gap

Table 16: From tasks to jobs. Share of employment (153.4 million jobs) in occupations whose own coverage measure is at least the threshold. Physical tasks, and every task of an occupation that cannot be done remotely (Dingel and Neiman, 2020) or needs close physical contact, count against the occupation (probability zero); the non-physical measure drops them from the occupation’s denominator. Geometric mean: CES aggregator of task success probabilities with substitution parameter $\rho \rightarrow 0$, weighted by task value. Weakest link: the lowest success probability among the occupation’s tasks with importance at or above its median. Observed use is not conditioned on.

Occupation-level measure	At least	Claude Opus 4.6	Mythos Preview (early)
Expected-success share of the occupation’s task value (additive)	50%	29%	29%
	75%	0%	8%
	90%	0%	0%
Share of the occupation’s task value shorter than the 80% horizon	50%	0%	5%
	75%	0%	0%
	90%	0%	0%
Share of the occupation’s task value shorter than the 50% horizon	50%	30%	30%
	75%	29%	29%
	90%	20%	25%
Expected-success share of the occupation’s <i>non-physical</i> task value	50%	37%	37%
	75%	0%	19%
	90%	0%	0%
Geometric mean of task success (CES, $\rho \rightarrow 0$)	50%	20%	20%
	75%	0%	6%
	90%	0%	0%
Weakest link: lowest success probability among above-median-importance tasks	50%	14%	20%
	75%	0%	1%
	90%	0%	0%

Table 17: Capability and observed use at each AEI data window. Frontier = longest 50 per cent horizon among models released by the window’s last day. Denominator: all tasks. “< p80” = share of the wage bill in non-physical tasks shorter than the frontier model’s 80 per cent horizon.

AEI window	Frontier model	Released	p50 (h)	p80 (h)	Expected success	< p80	Observed, window	Observed, cumulative
Dec 24–Jan 25	o1	5 Dec 2024	0.6	0.1	17%	0%	27%	27%
Feb–Mar 2025	Claude 3.7 Sonnet	24 Feb 2025	1.0	0.2	21%	0%	25%	30%
Aug 2025	GPT-5	7 Aug 2025	3.4	0.6	38%	3%	21%	32%
Nov 2025	Gemini 3 Pro	18 Nov 2025	3.7	0.9	40%	7%	26%	34%
Feb 2026	Claude Opus 4.6	5 Feb 2026	12.0	1.2	52%	9%	27%	34%
Apr 2026	Mythos Preview (early)	7 Apr 2026	17.4	3.1	60%	31%	23%	43%
May 2026	Mythos Preview (early)	7 Apr 2026	17.4	3.1	60%	31%	25%	44%

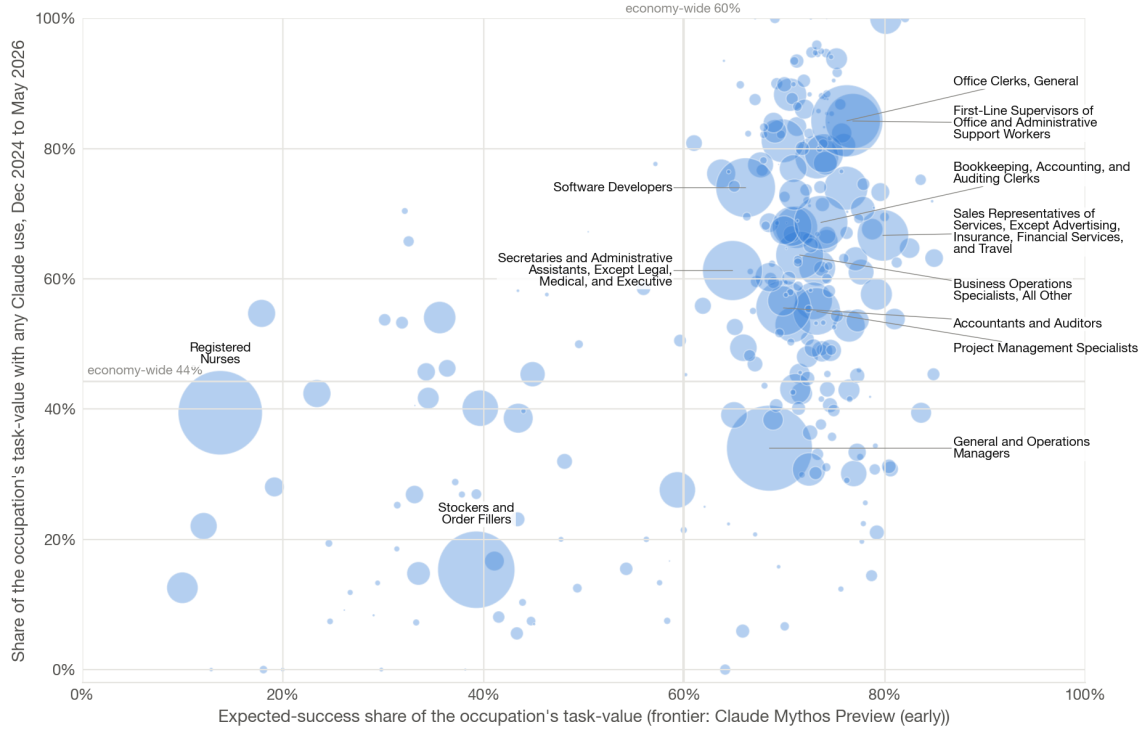


Figure 10: One bubble per 6-digit occupation that can be done remotely and without close contact (285 occupations), area proportional to employment. Horizontal axis: expected-success share of the occupation’s task value under Mythos Preview (early), physical tasks counting as zero. Vertical axis: share of the occupation’s task value with a published Claude.ai usage share in any AEI release, December 2024 to May 2026. Grey lines: economy-wide values on the baseline filter. The 12 largest occupations shown are labelled.

between two standards acts like a change in task length, with METR’s slope held fixed. If part of the gap is a quality standard that short tasks fail as well, a shift in the level of the curve would fit better and would penalise short tasks less. For the Remote Labor Index, the implied factor ranges from 52 to 330 across models, the largest comes from the one entry run on a different scaffold, each factor rests on a handful of accepted projects (Table 19), and the same factor is carried forward to a later model. For the AEI usage horizons, the curves I fit to the AEI’s own success labels do not identify a 50 per cent horizon on Claude.ai and place it at to hours on the API, against 3.5 hours in Appel et al. (2026) (Appendix I). Those rows combine a published ratio of horizons with METR’s slope, which the labels do not show.

I AEI Task Success and Task Length

The January and March 2026 AEI releases report, for each O*NET task and platform, the number of conversations that Claude classified as successful or unsuccessful, alongside the mean human time of those conversations. I keep tasks with at least ten classified conversations, which covers about 0.8 million conversations per window and platform, and regress success on $\log_2$ of human minutes in a conversation-weighted binomial model, pooled and with occupation or intermediate work activity fixed effects (Table 20). Figure 13 plots success by duration bin against the METR curves of the Claude models that bracket each window. For tasks under 15 minutes the curves predict 89 to 96 per cent success against 58 to 74 per cent observed; for tasks of four to sixteen

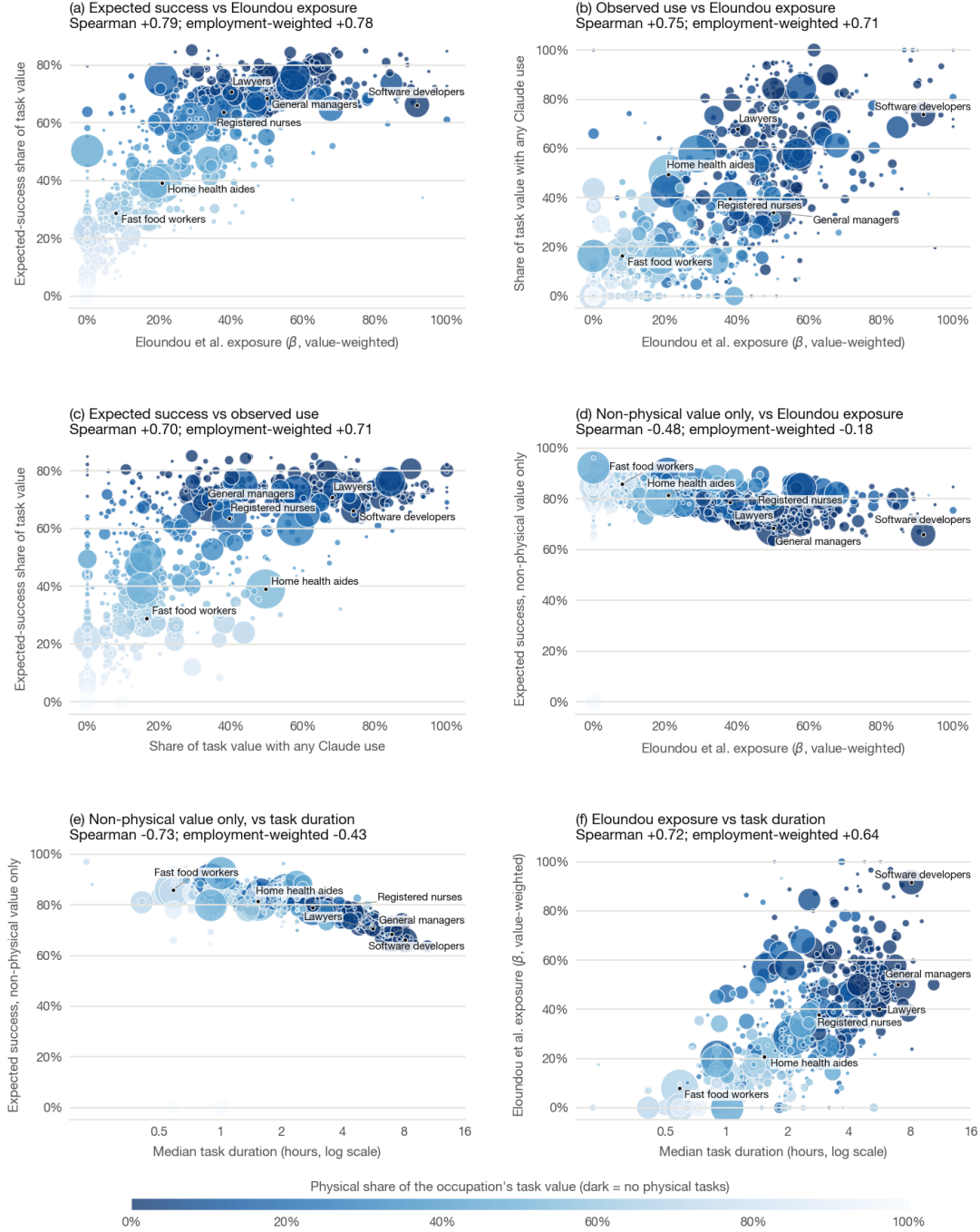


Figure 11: The comparison with rubric-based exposure measures described in Section 5.2, on the baseline filter. One bubble per 6-digit occupation, area proportional to employment, shade the physical share of the occupation's task value. Each panel prints the unweighted Spearman rank correlation across the 782 occupations and its employment-weighted Pearson counterpart. Panels (a) to (c) use the additive expected-success share, in which physical tasks enter at probability zero. Panels (d) and (e) use the expected-success share of the occupation's non-physical task value only, which is a function of task duration alone. Panel (f) plots exposure against duration. Eloundou exposure is the value-weighted mean of the task-level β labels within the occupation; frontier = Mythos Preview (early).

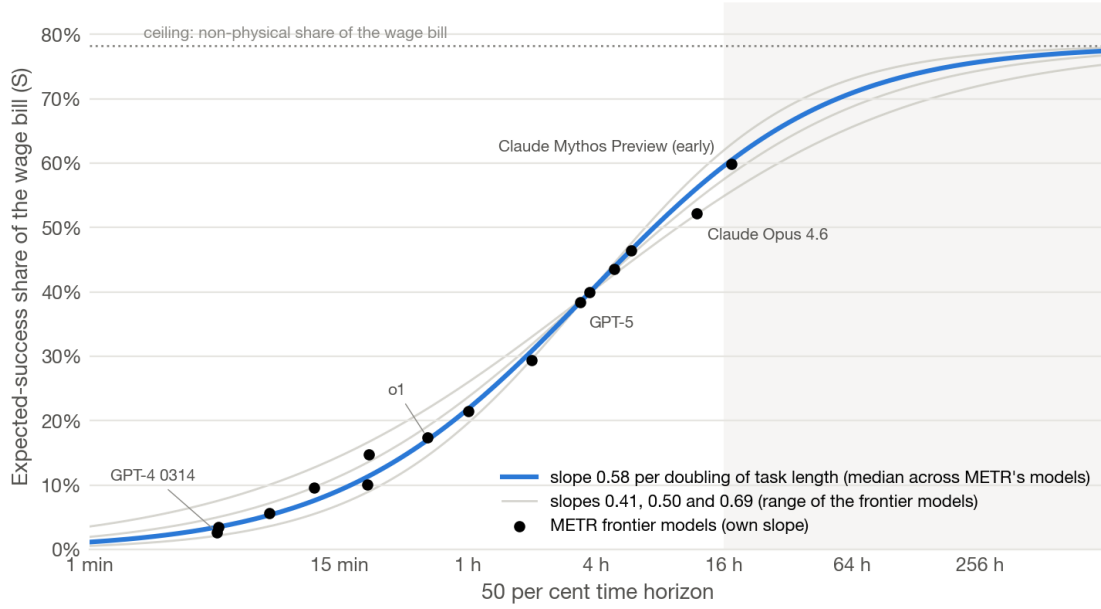


Figure 12: The expected-success share of the wage bill as a function of the 50 per cent time horizon, for four values of the slope of METR’s success curve. The slope is the fall in the log-odds of success for each doubling of task length: near the 50 per cent horizon, one doubling lowers the probability of success by about a quarter of the slope, 10 to 17 percentage points across these values. It also sets how far below the 50 per cent horizon the 80 per cent horizon lies: a quarter of it at a slope of 0.69 (o3), a tenth at 0.41 (Claude Opus 4.6). The four values span the frontier models, and each is one frontier model’s own slope. Points: METR’s frontier models at their own horizon and slope. Grey band: horizons METR describes as unreliable. Dotted line: the ceiling set by the non-physical share of the wage bill.

Table 18: The expected-success share under alternative success concepts, as illustrations (see text). Each row divides the frontier model’s 50 per cent horizon by the factor shown, holding its slope; months = $\log_2$ of the factor times the doubling time of our fit (4.2 months). †: derived from published figures rather than from data (Appendix A). The Remote Labor Index rows are the smallest and largest factors across the models in Table 19.

Calibration of the success concept	Horizon divided by	Months of progress	Expected success	
			Opus 4.6	Mythos
AEI usage, Claude.ai [†]	0.11	−14	69%	74%
AEI usage, API [†]	0.57	−3	57%	65%
GDPval, wins or ties against experts [†]	0.69	−2	56%	64%
Baseline: METR’s success concept	1	0	52%	60%
All durations doubled	2.0	+4	45%	52%
SWE-bench Verified (METR cross-domain)	3.1	+7	40%	46%
WebArena (METR cross-domain)	21	+18	21%	20%
Computer use, 40× shorter (METR) [†]	40	+23	16%	14%
Remote Labor Index, smallest factor [†]	52	+24	14%	12%
OSWorld (METR cross-domain)	66	+26	12%	10%
Computer use, 100× shorter (METR) [†]	100	+28	10%	8%
Remote Labor Index, largest factor [†]	330	+35	5%	3%

Table 19: Remote Labor Index automation rates against METR’s curves for the same model. Observed rate: share of 240 projects whose AI deliverable evaluators judged acceptable to a client (Scale AI leaderboard, accessed 24 September 2026), with a Clopper–Pearson interval. Predicted: expected success under METR’s curve over log-normal project durations with the published mean and median. Implied factor: the division of the 50 per cent horizon at which the curve reproduces the observed rate. The Claude Opus 4.6 entry used Anthropic’s Cowork scaffold rather than METR’s. Last row: GDPval’s win-or-tie rate for Claude Opus 4.1 at 7-hour tasks.

Leaderboard entry	METR curve	Observed (95% CI)	METR predicts	Implied factor	Months
claude-opus-4-6 (CoWork)	Claude Opus 4.6	4.2% (2.0–7.5)	51%	330	35
claude-opus-4-5-20251101-thinking	Claude Opus 4.5	3.8% (1.7–7.0)	37%	52	24
gpt-5.2-2025-12-11 (medium)	GPT-5.2	2.5% (0.9–5.4)	39%	86	27
gpt-5-2025-08-07	GPT-5	1.7% (0.5–4.2)	31%	81	27
gemini-3-pro-preview	Gemini 3 Pro	1.2% (0.3–3.6)	30%	67	26
GDPval, Claude Opus 4.1	Claude 4.1 Opus	47.6%	41%	0.69	–2

hours they predict less success than observed on Claude.ai. Across occupations, the rank correlation between the expected-success share of non-physical value and observed success is 0.05 to 0.26.

Using the pooled AEI curves in place of METR’s would not raise the headline share. On the baseline task frame, the API curves give an expected-success share of to per cent and the nearly flat Claude.ai curves to per cent, against 60 per cent under Mythos Preview (early). A flatter gradient raises success on long tasks but lowers it on short ones, and these curves describe the models that served the traffic, not the latest frontier. Against Claude Opus 4.6, the latest of those models (52 per cent), the API curves give less and the Claude.ai curves slightly more. Neither is an alternative estimate, because the labels measure success given delegation rather than autonomous completion.

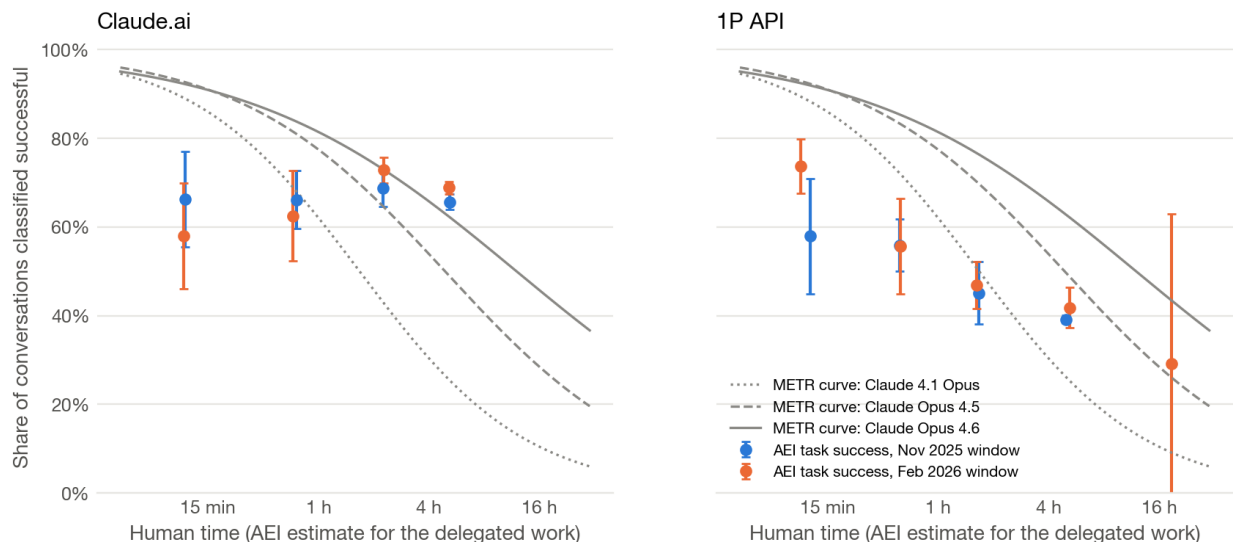


Figure 13: Share of conversations classified as successful by the AEI, by the task’s mean human time, for tasks with at least ten classified conversations; bars are 95 per cent intervals clustered by task. Lines: METR’s fitted curves for three Claude models. Left: Claude.ai. Right: API. The AEI labels are Claude’s classification of whether a conversation achieved what the user asked; they are conditional on delegation and allow iteration.

Table 20: AEI task success and human time. Binomial regressions of classified success on $\log_2$ human minutes, weighted by conversations; heteroskedasticity-robust standard errors in parentheses. The slope is reported as the fall in the log-odds of success per doubling, the same units as METR’s slopes (0.41 to 0.66 for the Claude models serving these windows). Within occupation: 6-digit SOC fixed effects. Within activity: fixed effects for the intermediate work activity of the task’s first detailed work activity.

Platform	Window	Tasks	Mean success	Fall in log-odds of success per doubling		
				Pooled	Within occupation	Within activity
Claude.ai	Nov 2025	2,388	68%	0.04 (0.05)	0.10 (0.03)	0.05 (0.04)
Claude.ai	Feb 2026	2,551	71%	−0.04 (0.07)	0.02 (0.03)	−0.06 (0.04)
API	Nov 2025	1,666	49%	0.19 (0.07)	0.12 (0.05)	0.17 (0.05)
API	Feb 2026	1,680	50%	0.19 (0.09)	0.13 (0.05)	0.21 (0.04)

J In-Person Rules

The Dingel–Neiman rule classifies an occupation as teleworkable unless any of fifteen conditions holds, such as rarely using email, working outdoors every day, or rating physical activities, handling objects, dealing with the public or inspecting equipment as very important (Dingel and Neiman, 2020). I transcribe the conditions from their published code and apply them to the Work Context and Work Activities files of O*NET 30.2. On the 720 occupation codes the two O*NET releases share, the rebuilt classification agrees with theirs on 94.6 per cent; the rest reflects revised O*NET ratings. On this paper’s weights, teleworkable occupations hold 51 per cent of the wage bill and 39 per cent of employment, against the 46 and 37 per cent that Dingel and Neiman report for 2018. Tasks in occupations without Work Context data (3.7 per cent of the wage bill) are kept. Table 21 gives the top ten occupations on the baseline filter, which the Dingel–Neiman rule removes from the top of the ranking. For the job-level results I add one condition that their code computes but leaves out of the rule: an O*NET physical-proximity rating (Work Context 4.C.2.a.3) below 4. It excludes school teachers, teaching assistants, childcare workers, food-service supervisors, ushers and gambling dealers, among others; a few exclusions are debatable, such as tutors and air traffic controllers. A threshold of 4.5 gives similar results (Table 13).

K Duration Channel and Data Vintage

Figure 14 shows two properties of the AEI duration that bear on its use as a task length. For the same task in the same window, API durations are 0.26 to 0.50 times Claude.ai durations, and shorter for 91 to 97 per cent of tasks; building the duration axis from API data instead gives an expected-success share of 70 per cent under Mythos Preview (early). Within the same task, Claude.ai durations barely change from November 2025 to February 2026 (ratio 1.00) or from April to May 2026 (1.06), but rise by a factor of 1.6 between February and April, when the AEI’s Claude.ai data first include Cowork. API durations do not rise over the same step (0.94). The rise is largest for tasks whose conversations produce code.

L Checking Costs

Suppose that delegating a task to the agent costs the worker checking time vD_t , that a detected failure means the worker redoes the task, and that an undetected or costly failure adds a further cD_t

Table 21: As Table 5, on the baseline filter (only physical tasks count as zero).

SOC	Occupation	Employment (thousands)	Expected success	Weakest link	Observed use	Median duration (h)
39-3091	Amusement and Recreation Attendants	398	85%	75%	29%	1.8
43-3011	Bill and Account Collectors	159	85%	72%	63%	1.7
45-4023	Log Graders and Scalers	3	85%	79%	0%	1.4
43-4151	Order Clerks	75	85%	74%	45%	1.7
43-4021	Correspondence Clerks	4	85%	74%	72%	1.7
45-2011	Agricultural Inspectors	14	84%	83%	51%	2.4
43-5061	Production, Planning, and Expediting Clerks	390	84%	78%	44%	1.7
43-5032	Dispatchers, Except Police, Fire, and Ambulance	203	84%	73%	39%	2.1
41-9041	Telemarketers	58	84%	80%	75%	2.7
13-2081	Tax Examiners and Collectors, and Revenue Agents	57	83%	86%	33%	1.8
31-1120	Home Health and Personal Care Aides	4,306	39%	0%	50%	1.5
41-2031	Retail Salespersons	3,898	60%	0%	58%	2.1
35-3023	Fast Food and Counter Workers	3,854	29%	0%	16%	0.6
11-1021	General and Operations Managers	3,503	69%	62%	34%	7.0
29-1141	Registered Nurses	3,380	64%	0%	39%	2.9

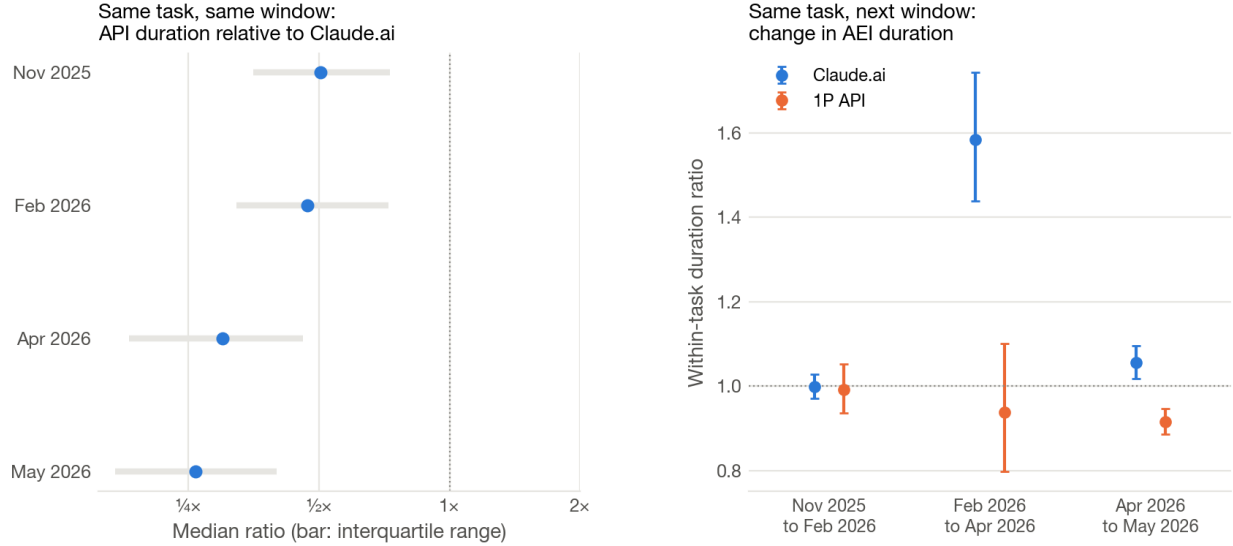


Figure 14: Left: for tasks observed on both platforms in the same window, the median ratio of the API duration to the Claude.ai duration (bar: interquartile range). Right: the wage-weighted mean within-task ratio of the duration in one window to the duration in the previous window, with 95 per cent intervals.

with probability $1 - P_t$. The expected time saved on a delegated task is then $D_t[P_t - v - c(1 - P_t)]$, and a worker delegates only when this is positive. Table 22 reports the resulting net share of the wage bill saved, for which $v = c = 0$ gives the expected-success share. Anthropic’s reliability adjustment of its own productivity estimate, from 1.8 to 1.0–1.2 percentage points a year (Appel et al., 2026), corresponds to v of 0.22 to 0.34 and a net share of 29 to 40 per cent. Only 8.8 per cent of the wage bill is in tasks with $P \geq 0.9$ under Mythos Preview (early), and 1.5 per cent with $P \geq 0.95$.

Table 22: Net share of the wage bill saved after checking costs. v : checking time as a share of the task’s human time. c : additional cost of a failure as a share of the task’s human time ($c = 0$: failures are caught and redone; $c = 1$: each failure costs one further task). Non-physical tasks only; physical tasks count as zero.

Checking time v (share of the task’s human time)	Mythos Preview (early)		Claude Opus 4.6	
	$c = 0$	$c = 1$	$c = 0$	$c = 1$
0.00	60%	41%	52%	26%
0.10	52%	34%	44%	19%
0.25	40%	23%	33%	10%
0.50	21%	8%	13%	2%